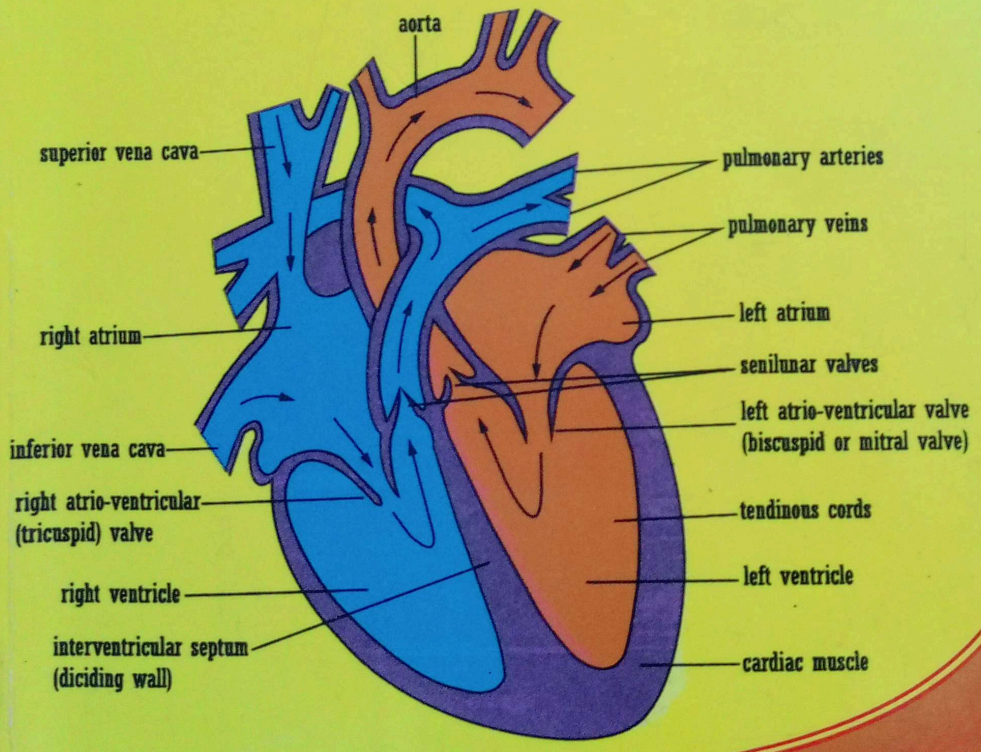
Extreme Series

# BIOLOGY

## GRADE 9-10




Extreme Series

2015 E.C

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Unit – 1

# 1. BIOLOGY AND TECHNOLOGY

**What is Biology?**

**Biology** is a branch of natural science which studies about living things.

Biological science is a multidisciplinary study which has various sub-branches of study. Among these, some of them include:

**Zoology** is the branch of biology which studies about animals.

**Botany** is the branch of biology which studies about plants.

**Taxonomy** is the study of classification and diversity of organisms.

**Anatomy** is the study of internal structure of an organism's body.

**Physiology** is the study of functioning of internal structure of living things

**Ornitology** is the study of birds.

**Microbiology** is the study of microorganisms

**Parasitology** is the study of parasites.

**Note:** Parasites are organisms that depend on living tissues.

**Entomology** is the study of insects.

**Mycology** is the study of Fungus.

**Fishery** is the study of Fishes.

**Scientific Method**

**Scientific Method** is a method that is used by scientists to solve problems in the society. The steps that are followed in the scientific method are:

1. Identifying the problem (asking the question) the problem can be identified through direct and indirect ways. Directly through observation, life experiences, exposure and indirectly through readying, leaning, internet, media broadcasts, etc.
2. Stating the hypothesis.
   **Hypothesis** is a tentative idea or statement that explains the observed facts and predicts the new outcomes.

3. Device ways to test the hypothesis by making observations, developing models and carrying out experiments.
4. Repeat the tests as often as necessary to determine weither the results will be consistent and as predicted.
5. Report the results of the tests and draw conclusions.

If the hypothesis is accepted, it is published in scientific publications which are called journals. Therefore, biologists publish their new ideas or works in scientific journals

**Peer review** is the process in which the published works of scientists (biologists) evaluated by others who are working in the same or related area.

Therefore, being a biologist is very interesting. There are various biologists at different parts of the world who made big contributions for the development of biological science. Among these, few of the most renowned ones are:

**Aristotle** – the first philosopher who also wrote intensively on biology (400 B.C). He is also one of the scientists who contributed a lot for the beginning of 'life studies' (biology) as a discipline,

**Robert Hooke** – (1665) – is responsible for the discovery of cells.

**Charles Darwin** – the father of evolution.

**Louis Pasteur** – the first person who disproved the theory of spontaneous generation and formulated the germ theory of disease.

**Gregor Mendel** – the father of modern genetics.

**Thodre Schwann and Mathias J.scheilden** – formulated the cell theory.

**Edward Jenner** – developed the vaccination against small pox.

**Rachel Carson** – the first environmentalist.

**James Watson, Fransis Crick, Maurice Wi/kins** and **Rosiland Franklin** introduced the model of the DNA molecule which is also referred to as 'the seret of life:

**Illustrative Example**

1. In scientific method which one comes at last?
   A. Identification of the problem
   B. Testing the hypothesis
   C. Drawing conclusions
   D. Formulating hypothesis

***Explanation***: The correct step in scientific method is

1) Indentifying the problem
2) Formulation of hypothesis
3) Testing the hypothesis
4) Drawing conclusions
5) Reporting the results

**Answer: C**

2. The process in which published works of biologists is evaluated by others in the same field is called:

A. Hypothesis
B. Journal
C. Peer review
D. Report

***Explanation***: Peer review is a very common aspect among scholars in which the publication of one author or scientists is criticized or reviewed by others in the same field.

**Answer: C**

3. Among the following renowned biologists which one is the pioneer?

A. Anton Van Leuwenhoek
B. Charles Darwin
C. Aristotle
D. Louis Pasteur

***Explanation***: Aristotle is the pioneer in the Field of science. Even though his name is very popular in different fields of stud he was lived 400B.C

**Answer: C**

4. Which of the following biologists is incorrectly matched with their works?

A. Louis Pasteur – Disproved theory of spontaneous generation
B. Aristotle – formulated germ theory of disease
C. Gregor Mendel – the father of genetics
D. Charles Darwin – The Father of evolution

***Explanation***: The germ theory of disease is formulated by biologists like **Louis Pasteur and Robert Koch**. Aristotle is known for his theory of spontaneous generation which states, certain non living things can be changed in to a living thing.

**Answer: B**

5. Among the following biologists which ones is known for the formulation of cell theory?

A. Louis Pasteur
B. Theodre Schwann
C. Gregor mendel
D. Lazoro spatanzani

***Explanation***: The cell theory is formulated by the following scientists mainly:

- Rudolf Virchow
- Theodre Schwann
- Mathias Jackob schielden

**Answer: B**

## Post Test 1

**Part I: Choose the best answer from the given alternatives.**

1. In Scientific method which one comes first?
   A. Testing the hypothesis
   B. Formulating hypothesis
   C. Identification of the problem
   D. Drawing conclusions
2. A tentative idea or statement that explains observed facts and predicts new outcomes is:
   A. Journal C. Report
   B. Hypothesis D. Peer review
3. Evaluation of published works of scientists by others in the same area is:
   A. Assessment C. Peer counseling
   B. Peer review D. Evaluation
4. Which one of the following scientists is correctly associated with his contribution?
   A. Aristotle: invented microscope
   B. Anton Van Leeuwenhoek: discovered cells
   C. Gregor Mendel: explained the mechanism of inheritance
   D. Charles Darwin: the father of genetics
5. The study of insects is:
   A. Entomology C. Fishery
   B. Parasitology D. Microbiology

## 1.1 Renowned Ethiopian

Even though our country Ethiopia has various renowned biologists who made great contribution in different fields of biological science, some of them will be discussed as follows:



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## Professor Aklilu Lemma

- Professor Aklilu Lemma is the pioneer in the development of biological science in Ethiopia. Among so many contributions of him the massive one is his work on the battle against bilharzia (schistomiasis).
- Bilharzia is a parasitic worm which parasitizes humans. It affects more than 300 million people in tropical areas of Africa, South America, Asia and parts of the Caribbean.
- The bilharzia worm uses **snail** as intermediate host in which it spends some part of its life cycle.
- Professor Aklilu Lemma made soup berry from a plant which is commonly called **Endod** in Ethiopia or ***Phytolecca dodecandria,*** is used to eradicate Snails which are intermediate hosts of bilharzia worm. This huge discovery is made based on the observation which Professor Aklilu Lemma made around Adwa in 1964.
- The soup berry of Proff. Aklilu Lemma is advantageous over the **molluscicide** chemicals because it is cheaper and affordable for more Africans to buy and use it.
- Based on his discoveries Professor Aklilu Lemma recognized his discovery by saying "We found a poor man's medicine for poor man's disease."

## Doctor Tewolde Berhan GebreEgziabher (The ardent lover of nature)

Dr. Tewolde Berhan is devoted to work in the following major areas:

- Exemplary work to safeguard the biodiversity
- Traditional rights of farmers and communities to their genetic resources.

- Dr. Tewolde Berhan also participated in various biodiversity meetings and negotiations. Some of these include:
  - Biodiversity related meetings such as, Convention on Biological Diversity (CBD) and the food and Agriculture Organization.
  - Chief negotiator on bio safety for likeminded group (G – 77 countries).
- For his massive contributions, Dr. Tewolde Berhan have Won various international awards. Among these some of them are:

- The right lively award by the year 2000. (Also called alternative Nobel Prize).
- United Nations top environmental prize, champions of the Earth in 2006.

## Professor Tilahun Yilma

Professor Tilahun Yilma is highly recognized for his vaccines against terrible cattle disease called **rindpest**. These vaccines are made through quite a modern technique called **genetic engineering**. The vaccine is simply scratched on the neck or abdomen of cattle without any form of injection.

Professor Tilahun is also working for his work on HIV/AIDS vaccine. He is also highly devoted to produce high quality researches and encouraging young scientists.

**Professor Yealemtsehay Mekonnen**

Professor Yealemtsehay is the first Female Professor in Ethiopia From Addis Ababa University. She has worked in various areas. Among these some are:

- Effect of chemical pesticide hazard on humans in Awash agricultural farms.
- Research on the use of plants as medicine against human and animal diseases.

Professor Yealemtsehay also participated and contributed as:

- Head of Biology Department at Addis Ababa University,
- Director of Aklilu Lemma foundation.
- President of Biological society of Ethiopia.

Professor Yealemtsehay awarded research grants and fellowships from.

- Ethiopian science and technology commission.
- Ethiopian Agricultural Research organization.
- British council.
- International foundation for science.
- Third world Academy of sciences.
- Alexander Von Humboldt foundation.

## Dr. Melaku Worede

Dr. Melaku Worede is an Ethiopian plant genetic researcher. Among various contributions of him in relation to plant genetic resources of Ethiopia, some of the major ones are:

- To save genetic diversity of Ethiopian domestic plants.
- Preserved traditional crop varieties.
- He worked on how to get high yield of crops without using commercial fertilizers.

Dr. Melaku also worked as a chair of United Nations Food and Agriculture Organization commission on Plant genetic resources.

For his works and contributions Dr. Melaku awarded:

- Right Lively award in 1989.
- National Green Awarded Foundation for his outstanding international contributions.

## Dr. Gebissa Ejeta

Dr. Gebissa is highly recognized for his work on developing stains of sorghum which are resistant to drought and produce high crop yield which is up to ten times. He also developed strains of sorghum which are resistant to parasitic **striga weed**.

Accordingly, the contribution of Doctor Gebissa has made a massive difference in sustaining the food security in Ethiopia and other African counties.

For his massive contribution in science and technology Dr. Gebissa awarded the **'National hero award of Ethiopia'**

## Professor Beyene Petros

Professor Beyene is a biomedical scientist from Addis Ababa University. Among various contributions made by professor Beyene, some are:

- He produced more than 43 publications in Journals and books
- He worked as chair person and advisory committee on Health Research and Development, WHO/AFRO, the vice minister of Education and other scientific societies.

For his huge contribution professor Beyene have won:

- Gold medal award from Ethiopian Health Association.
- Fellowships from Fulbright and from centers for disease control and prevention, Atlanta, USA.

## Professor Sebsibe Demisew

Professor Sebsebe is a plant taxonomist. He is currently working as:

- The director of the National Herbarium.
- The leader of Ethiopian Flora Project (EFP).

## Dr. Zeresenay Alemseged

Dr. Zeresenay is known for his discovery of a 3.3 million years old humanoid fossil in 2006.

- *The study of fossil is called* ***paleontology***

## Dr. Tsehaynesh Meselle

Among various contribution of Dr. Tsehaynesh, some are:

- She worked as a Director of Ethiopian Health and Nutrition Research Institute (EHNRI)
- She wrote books and conducted research on human health issues, including HIV/AIDS.

## Dr. Berhane Asfaw

Dr. Birhane is an evolutionist who discovered two 160,000 year old human skulls which have a big impact in the study of human evolution.

## Professor Legesse Negash

Pro. Legesse is a plant physiologist from Addis Ababa University who has made various contributions. Among these some are:

- He is the pioneer in propagation of Ethiopian indigenous trees.



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➢ He is the founder and leader of the center for indigenous trees propagation and biodiversity development in Ethiopia.

For his massive contributions professor Legesse have won international awards including the one from Stockholm – based International Foundation for science.

- **Professor Mogessie Ashenafi** is an international researcher in to food microbiology.
- **Professor Ensermu Kelbessa** is a systematic botanist who discovered and named various new plants.

**Illustrative Example**

6. Among the following list of renowned Ethiopian biologists which one is considered to be the pioneer?
   A. Professor Yealemtsehay Mokonnen
   B. Professor Aklilu Lemma
   C. Dr. Tewoldeberhan G/Egziabher
   D. Dr. Gebissa Ejeta

   ***Explanation***: Professor Aklilu Lemma made a great contribution by making a soup berry which eradicates snails which are intermediate hosts of bilharzia in 1964. **Answer: B**

7. Ethiopian Biologist who worked to increase the crop productivity of Ethiopia is:
   A. Professor Beyene Petros
   B. Dr. Gebissa Ejeta
   C. Professor Aklilu Lemma
   D. Professor Tilahun Yilma

   ***Explanation***:

   Dr. Gebissa Ejeta worked to develop strains of sorghum which are resistant to drought and resistant to striga weed. He also introduced his works can increase the crop productivity of Ethiopia by ten times. **Answer: B**

8. What is the role of soap berry of Professor Aklilu Lemma in controlling bilharzias?
   A. Killing the bilharzias worms inside the body of human.
   B. Eradication of snail which is an intermediate host for bilharzia.
   C. Cleaning the clothes by the soup made from Endod.
   D. Removal of stagnant water bodies where the snails breed,

   **Answer: B**

9. Ethiopian biologist who is known for his vaccines against rind pest is:
   A. Dr. Tewoldeberhan G/Egziabher
   B. Professor Tilahun Yilma
   C. Dr. Melaku Worede
   D. Dr. Gebissa Ejeta

   *Explanation*: Professor Tilahun Yilma is known for:
   - His vaccines against terrible cattle disease rind pest
   - Works on the vaccines of HIV/AIDS

   **Answer: B**

10. Ethiopian biologist who worked on the effect of chemical pesticides on human is:
   A. Professor Yealemtsehay Mekonnen
   B. Professor Beyene Petros
   C. Professor Tilahum Yilma
   D. Dr. Melaku Worede

   *Explanation*: Professor Yealemtsehay Mekonnen, who is the first female professor from A.A.U, works on the following major areas:
   - The effect of chemical pesticides on humans
   - The use of plants for medicinal source.

   **Answer: A**

11. Which of the following Ethiopian biologist awarded the National Hero Award of Ethiopia?
   A. Professor Sebsibie Demisew
   B. Dr. Zeresenay Alemseged
   C. Dr. Tsehaynesh Meselle
   D. Dr. Gebissa Ejetta

   *Explanation*: Dr. Gebissa Ejetta awarded the National Hero award of Ethiopia for his works on sorghum. **Answer: D**

12. Ethiopian Evolutionist who discovered old human skulls which have a big impact on human evolution:
   A. Professor Legesse Negash
   B. Professor Mogessie Ashenafi
   C. Professor Ensermui Kelbessa
   D. Dr. Berhane Asfaw

13. Why a hypothesis is called 'an educated gues'? **Answer: D**

   *Explanation*: The hypothesis is a tentative explanation or idea which is formulated based on some background knowledge or information about the problem encountered,

14. Why is Professor Aklilu Lemmas work on the battle against bilharzias so important for most Africans?

***Explanation***: Professor Aklilu Lemmas contribution is highly valuable for most African because before the discovery of Prof. Aklilu soapberry. The other molluscides in use were very expensive. Thus, Dr. Aklilu's work soup berry is cheap and affordable by most Africans.

15. What makes the vaccine of Dr. Tilahun Yilma against find pest, so special?

***Explanation***:

The vaccine against terrible cattle disease rind pest which is made by Dr. Tilahun Yilma is so special because:

- It is made through a modern application of genetic engineering.
- It is easy to use.

16. Among the contributions of Dr. Melaku worede list of three of them.

***Explanation***:

Dr. Melaku worede is highly accredited for his works:

- To save genetic diversity of Ethiopian domestic plants
- To preserve traditional crop varities.
- Ways of getting high yield of crops without using commercial fertilizers.

17. Among the most renowned Ethiopian biologist, who has been honored by developing high yielding strains of sorghum? (EGSECE, 2005 E.C)

A. Dr. Aklilu Lemma
B. Prof. Beyene Petros
C. Dr. Gebissa Ejeta
D. Prof. Tilahun Yilma

***Explanation***:

| **Renowned Ethiopian biologist** | **Contribution** |
|---|---|
| Prof. Aklilu Lemma | He made a soup berry to eradicate snail, which is intermediate host for bilharzia. |
| Prof. Beyene Petros | He is a biomedical scientist |
| Dr. Gebissa Ejetta | Produced high yielding and drought resistant variety of sorghum |
| Prof. Tilahun Yilma | Produced vaccine against terrible cattle disease called **rindpest** |

**Answer: C**

## *Post Test 2*

***Part I:*** **Choose the best answer from the given alternatives.**

1. A renowned Ethiopian biologist who worked on increasing the productivity of sorghum is:
   A. Dr. Aklilu Lemma
   B. Pro. Mogese Ashenafi
   C. Pro. Gebisa Ejeta
   D. Pro. Beyene Petros
2. Aklilu Lemma contributed in prevention of a disease called:
   A. Malaria
   B. Sleeping sickness
   C. Bilharzia
   D. Marasmus
3. Vaccine against terrible cattle disease rind pest is produced by:
   A. Pro. Yealemtsehay Mekonnen
   B. Dr. Aklilu Lemma
   C. Pr. Tilahun Yilma
   D. Dr. Birhane Asfaw
4. Ethiopian paleontologist who discovered 3.3 million years old humanoid fossil is:
   A. Dr. Zeresenay Alemseged
   B. Pr. Tilahun Yilma
   C. Pr. Hinsermu Kelbesa
   D. Pr. Mogese Ashenafi
5. The use of indigenous plants to be used as a source of medicine is studied by :
   A. Dr. Tsehaynesh Mesele
   B. Pr. Yealemtsehay Mekonnen
   C. Pr. Gebisa Ejeta
   D. Pr. Tilahun Yilma
6. Ethiopian biomedical scientist who have published various journals and books is:
   A. Pr. Beyene Petros
   B. Dr. Teweldeberhan G/Egzihabher
   C. Dr. Zeresenay Alemeged
   D. Pr. Yealemtsehay Mekonnen
7. An indigenous plant that is used to produce a moluscide to eradicate snail which is the intermediate host of bilharzias is:.
   A. Serbo
   B. Eucalyptus
   C. Endod
   D. Wanza
8. _____ is well renowned for his contribution on environmental protection.
   A. Pr. Tilahun Yilma
   B. Pr. Beyene Petros
   C. Dr. Tewoldeberhan G/Egzihabher
   D Pr. Gebisa Ejeta



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## 1.2 Biolo

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9. The first act of propagating Ethiopian indigenous trees is done by:
   A. Pr. Mogese Ashenafi
   B. Pr. Yealemtsehay Mekonnen
   C. Pr. Tlahun Yilma
   D. Pr. Legese Negash
10. An Ethiopian genetic Engineer who researched on HIV/AIDS Vaccine is:
   A. Pr. Yealemtsehay Mekonnen
   B. Dr. Melaku Worede
   C. Dr. Aklilu Lemma
   D. Pr. Tiahun Yilma

## 1.2 Biological research in Ethiopia

Ethiopia has number of institutions which are working in relation to biological research. However, a lot should be done in this area in order to make them internationally accepted and renowned.
Some of the institutions in Ethiopia which play an important role in the development of biological research in are:

### Addis Ababa University (Biology Department)

- Biology department which is located at faculty of science in Addis Ababa University plays a major role in the development of biology as a science in Ethiopia. Some of these are:
  - A center for Biological Research in Ethiopia.
  - Home for the Aklilu Lemma Institute of pathobiology
  - Consists of much more modern and high level equipments to help biologists in their research.
  - A work place for most Ethiopian renowned biologists.

### Armauer Hansen Research Institute (AHRI)

- AHRI is founded in 1969 around the place which is called Zenebe work in Addis Ababa.
- AHRI was highly dedicated to patients with Leprosy and research only to Leprosy. However, starting from 1996, AHRI diversified its areas of work in to **tuberculosis, Leshmaniasis, Malaria** and **HIV/AIDS**, in addition to their work on to leprosy.

## Aklilu Lemma Institute of Pathobiology (ALIPB)

- ALIPB is named on the honor of Professor Aklilu Lemma. It was previously known as Department of pathology at Addis Ababa University.
- ALIPB is the center for biomedical research and training.
- The major research areas of ALIPB is:
  1) Microbiology research program in to major infectious disease.
  2) Vectors of diseases and how to control them.
  3) Human parasitic diseases.
  4) Animal health disease.
  5) Research in to Endod and use of other plants as a source of medicine.
- ALIPB is also training new Ethiopian pathobiologists.

***Pathobiologist:*** is a biologist who studies about the causes of communicable diseases.

## Ethiopian Health and Nutritional Research Institute (EHNRI)

EHNRI is an institute that research in to health and nutrition in relation to public health issues.

The specified work areas of EHNRI include:

- Research in to immunology and viral disease though its well equipped laboratory.
- Works in relation to HIV/AIDS including trial on HIV vaccine.
- The battle against tuberculosis (TB).
- Nutritional states of mothers and babies.
- Infectious disease.

## Ethiopian Institute of Agricultural Research (EIAR or IAR)

involves cultivation of the land or livestock through biological research. Accordingly, IAR is highly devoted to research in to the following research areas.

- Development of crop technology to achieve food security and nutritional quality.
- Ways of managing animals (live stock). This involves breeding and feeding programmes and health improvement of our livestock.
- Soil and water. This involves ways of improving the fertility of the soil, specially by avoiding buying expensive inorganic fertilizers and effective watering of the land.
- Forestry which involves in rehabilitation, restoring and conserving the forest ecosystems.
- Mechanizing farming which involves mechanized harvesting.

## Institute of Biodiversity conservation (IBC)

➢ The issue of biodiversity conservation is an internationally recognized issue.

**Note:** Biodiversity refers to the diversity of all the life forms in an area.
IBC is highly dedicated to conserving the genes of Ethiopian endemic plants. Currently the institute is involved in conservation of plants, animals and microorganisms in Ethiopia. In addition, IBC is working on the management of the ecosystem.
Therefore, the major areas of research of IBC are:

- Forest
- Aquatic plants
- Medicinal plants
- Animal genetic resources
- Biotechnology and safety
- Ecosystem conservation

➢ The IBC founded one of the leading gene banks in Africa with over 300 plant species.

## Illustrative Examples

18. Ethiopian biological research center which is highly devoted on researches related to Leprosy is:
A. ALIPB B. AHRI C. EHNRI D. IAR

*Explanation*:
Armauer Hansen Research institute (AHRI) was devoted to issues related to Leprosy. However, currently AHRI is also working on TB, Leshmania, Malaria and HIV/AIDS

**Answer: B**

19. Which of the following is NOT the major work areas EHNRI?
A. Battle against TB
B. Improving nutritional state of mothers and babies.
C. Mechanized Farming
D. Trial on HIV/AIDS vaccine

*Explanation*:
The major areas of research of EHNRI is:
- Immunology and viral disease
- HIV vaccine trial
- Battle against TB
- Nutritional states of mothers and babies
- Infectious diseases

**Answer: C**

20. Ethiopian biological research center which is involved in development of crop technology to achieve food security and nutritional quality is:
A. EIAR B. EHNRI C. AHRI D. ALIPB

*Explanation*:
Among various research areas of EIAR is to achieve food security and nutrition quality is one.

**Answer: A**

21. Among the most renowned Ethiopian biologis, who was been honored by developing high yielding strains of sorghum? (EGSECE, 2005 E.C)
A. Dr. Aklilu Lemma
B. Prof. Beyene Petros
C. Dr. Gebissa Ejeta
D. Prof. Tilahun Yilma

*Explanation*:

| Renowned Ethiopian biologist | Contribution |
|---|---|
| Dr. Aklilu Lemma | Soap berry to eradicate snail, which is intermediate host for bilharzia. |
| Prof. Beyene Petros | Biomedical scientist |
| Dr. Gebissa Ejeta | Produced high Yeilding and drought resistant variety of sorghum. |
| Prof. Tilahun Yilma | Vaccine for terrible cattle disease called **rindpest.** |

**Answer: C**

22. Why is the establishment of biological research institutes so important in Ethiopia?

*Explanation*: Establishment of biological research institutes is important for our country Ethiopia; because:

- It helps to upgrade the standard of biological research.
- It helps to train highly skilled researchers and scientists.
- To increase international partnership with other countries.
- For the overall development of our country Ethiopia.

What are the major diseases which AHRI is working on?

*Explanation*: currently the AHRI is dedicated to work on diseases such as **Leprosy, tuberculosis, leshmaniasis, malaria** and **HIV/AIDS.**

23. List out at least two biological research centers in Ethiopia which are working on Tuberculosis (TB).

*Explanation*: Among various organizations, some of the Biological research institutes which are devoted to work on infectious diseases such as TB are BAHRI, ALIPB, EHNRI.

24. The Ethiopian which have given an emphasis to pathobiology is:

A. IBC B. ALIPB C. EHNRI D. AHRI

**Answer: B**

25. Among the following institutes, which one is researching on endod and other plants as a source of medicine?

A. EHNRI B. IBC C. IAR D. ALIPB

*Explanation*: ALIPB is a biological research institute which is highly devoted to research in to pathobiology including use of 'endod' and other plants as a source of medicine

**Answer: D**

26. Which of the following is NOT among the research areas of IAR?
   A. Development of crop technology
   B. Managing livestock
   C. Soil and water
   D. Vectors of diseases **Answer: D**

27. An institution that carries out research in to leprosy and tuberculosis is: (EGSECE, 2006 E.C)
   A. Ethiopian health and Nutritional Research Institute
   B. Ethiopian instate of agricultural research
   C. Armauer Hansen Research Institute.
   D. Institute of Biodiversity conservation

   ***Explanation***: Armauer Hansen Research institute is highly deveted to research in to the following diseases:
   - Tuberculosis
   - Malaria
   - HIV/AIDS
   - Leshmaniasis
   - Leprosy

   **Answer: C**

## *Post Test 3*

***Matching***

| Column A | Column B |
|---|---|
| 1. IBC | A. Development of crop technology to achieve food security and nutritional quality |
| 2. ALIPB | B. Research in to immunology and viral disease though its well equipped laboratory |
| 3. ENHRI | C. dedicated to patients with Leprosy and research only to Leprosy |
| 4. AHRI | D. Research in to Endod and use of other plants as a source of medicine. |
| 5. IAR | E. conserving the genes of Ethiopian endemic plants |

## Supplementary Exercise on Unit One

**I. Multiple choice**

1. What makes the soap berry of Professor Aklilu Lemma advantageous over other Molluscides?
   A. Its available in every drug store.
   B. Its relatively cheaper.
   C. It's quite a modern technology.
   D. Its supplied to every patient for free.

2. Morphology is the branch of biology which deals with:
   A. The functions of internal structures of the body.
   B. The internal structures of the body.
   C. The classification of organisms.
   D. The function of the external structures of the body.
3. A tentative idea or statement that explains observed facts and predicts new outcomes is:
   A. Journal
   B. Article
   C. Principle
   D. Hypothesis
4. A biologist who is very popular for his discovery of cells is:
   A. Antovan Leuweahock
   B. Aristotle
   C. Robert Koch
   D. Robert Hooke
5. A plant species from which professor Aklilu Lemma made a soap berry is:
   A. *Juniperus procera*
   B. *Rosa a bysinica*
   C. *Phytolecca dodecandra*
   D. *Juniperus abysinica*
6. An Ethiopian biologist who is known as the ardent lover of nature is:
   A. Dr. Gebisa Ejeta
   B. Dr. Tewoldeberhan G/Egziabher
   C. Professor Yealmtsehay Mekonnen
   D. Professor Tilahun Yilma
7. Ethiopian biologist who used a genetic engineering technology to develop vaccines is:
   A. Professor Tilahun Yilma
   B. Professor Yealemtsehay Mekonnen
   C. Dr. Melaku Worede
   D. Dr. Gebissa ejeta
8. Ethiopian plant genetic researcher who worked to save genetic diversity of Ethiopian domestic plants is:
   A. Professor Beyene Peteros
   B. Dr. Melaku worede
   C. Dr. Gebissa ejeta
   D. Dr. zeresenay Alemseged.
9. All of the following are correct about professor Beyene Petros, except.
   A. He is a biomedical scientist
   B. He published various articles

C. He won a gold medal award from Ethiopian Health Association
D. He is currently working as a director of National Herbarium.

10. An Ethiopian Female scientist who wrote books and conducted research on human health issues, including HIV/AIDS is:
A. Professor Yealemtsehay Mekonnen
B. Dr. Tsehaynesh Meselle
C. Dr. Bogetch Gebre
D. None

11. Professor Mogessie Ashenafi is:
A. A plant physiologist
B. Food microbiologist
C. Human evolutionist
D. Biomedical scientist

12. A biological research institute which is considered to be a center for Biological research in Ethiopia is:
A. Ethiopian Institute for Agricultural research
B. AAU biology department
C. Armauer Hansen research institute
D. Institute of Biodiversity conservation

13. All of the following are research areas of IBC, **except**
A. Forestry
B. Medicinal plants
C. biotechnology and safety
D. Battle against tuberculosis

14. A biological research institute which is researching into Endod and use of other plants as a source of Medicine is:
A. ALIPB
B. AHRI
C. EHNRI
D. IAR

15. Which of the following biological research institute is working on biodiversity related issues?
A. IAR
B. IBC
C. EHNRI
D. AHRI

16. The biological research institute which is working on preservation of endemic plants and animals of Ethiopia is:
A. IBC B. IAR C. AHRI D. AAU

17. All of the following are the research areas of ALIPB, except?
A. Human parasitic diseases.
B. Vectors of disease and how to control
C. Nutritional state of mothers and babies
D. Animal health and disease.



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Column

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27. Pr M
28. Pr P
29. P D
30. D

Answers

1. C

Answers

1. C
7. C

## II. Blank Space items

18. The first female Ethiopian professor from A.A.U. is _____
19. Scientific discoveries and articles of scientists are published in _____.
20. Professor Tilahun Yilma is highly recognized for his vaccines against terrible cattle disease called _____.
21. A systematic Ethiopian botanist who discovered and named various new plants is _____.
22. ALIPB is an abbreviation which stands for _____.
23. Ethiopian institute which is working on an internationally recognized issue of biodiversity is _____.
24. _____ is a condition in which the published works of scientists is evaluated by others in the same Field.
25. A tentative idea or statement that explains observed facts and predicts new outcomes is _____

## III. Matching

| Column A | Column B |
|---|---|
| 26. Professor Tilahun Yilma | A. a biomedical scientist |
| 27. Professor Yealemtsehay Mekonnen | B. plant taxonomist |
| 28. Professor Beyene Petros | C. Research on the use of plants as medicine against human and animal disease |
| 29. Professor sebsbie Demisew | D. a plant genetic researcher |
| 30. Dr. Melaku Worede | E. works on HIV/AIDS vaccine |

**Answers to Posttest I**

1. C 2. B 3. B 4. C 5. A

**Answers to Post test II**

1. C 2. C 3. C 4. A 5. B 6. A
7. C 8. C 9. D 10. B

**Answers to Post Test 3**

1. E 2. D 3. B 4. C 5. A

## Answers to Supplementary Exercises on Unit One

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | B | 2. | D | 3. | D | 4. | D | 5. | C |
| 6. | B | 7. | A | 8. | B | 9. | D | 10. | B |
| 11. | B | 12. | B | 13. | D | 14. | A | 15. | B |
| 16. | A | 17. | C | | | | | | |

18. Pr. Yealemtsehay Mekonnen
19. Journals
20. Rindpest
21. Pr. Ensermu Kelbessa
22. IBC
23. Peer review
24. Hypothesis
25. D 26. A
27. F 28. B
29. C 30. E

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Unit – 2

# CELL BIOLOGY

## 2.1. The microscope

**Microscope** is one of the basic tools that are used by biologists in order to study living things. It is a device which is used to view or study parts of an organism and very small organisms such as ***bacteria*** and ***protozoa***. Thus, since it is used largely in biological studies, the microscope is called the ***basic tool of a biologist***.

The first light microscope is invented by a Dutch biologist Called **Anton Van Leeuwenhoek**. Therefore, the discovery of microscope has largely improved our knowledge and understanding of cell and its structure.

**The abilities of Microscope**

The Microscope has two major abilities:

1) ***Magnification*** –is increasing the size of an object or specimen to be viewed. The best light microscopes can magnify up to 2000 times (2000X). However, electron Microscope can magnify up to two million times (2,000,000X). Thus, electron microscope helps us to study more about the cells and their functions.
2) ***Resolution*** – is the ability of the microscope to show the detail or the scattered parts of an object. It helps us to distinguish between two separate points. The measure of how much detail the microscope can show is called **resolving power.** Thus:
    - *the greater the resolving power of the microscope, the more detail it can show.*

- The resolving power and the wave length are inversely proportional. Thus, the higher the wavelength, the smaller the resolving power.
- The electron microscope has higher resolving power than the light microscopes.

## Types of Microscope

Basically there are two major types of microscopes; namely,

1) ***Light microscope*** – is a type of microscope that uses a beam of light to form the image of an object.
2) ***Electron microscope*** – is a type of microscope that uses beam of electrons to form an image. It is the most advanced type of microscope.

### The light microscope

The light microscopes are used to study the cells, tissues or even individual organisms.

The light microscopes can be of two types. These are:

1. **Simple light microscopes** – have only one lens system.
   **Examples**: Hand lens, Magnifying lens, reading lens.
2. **Compound light Microscope** – have two lens systems. These are eye piece (ocular) lens and objective lenses.
   The objective lenses include;
   - **Low power objective**
   - **Middle power objective**
   - **High power objective**
   - **Oil immersion lenses**

Therefore, in a compound light microscope, the total magnification power is the product of an eye piece and the objective lens used.



Total magn
power

parts
1 Eyepi
2 Objec
3 Stage
4 Cond
5 Illum
6 Base
7 Binoc
8 Nosep
9 Arm
10 Coa
11 Iris
12 Stag
13 Pow

Parts o

A compou
Base – sup
Mirror –

Fine adju

Diaphrag

Objective
Stage – su
Nose piec
Arm – sup
Course ad

Total magnification = magnification power × magnification power of the objective lens used
power of eye piece

**parts**

1 Eyepiece
2 Objective lenses
3 Stage holder clamp
4 Condenser
5 Illuminator
6 Base
7 Binocular head
8 Nosepiece
9 Arm
10 Coaxial focus knob
11 Iris diaphragm
12 Stage control knobs
13 Power switch

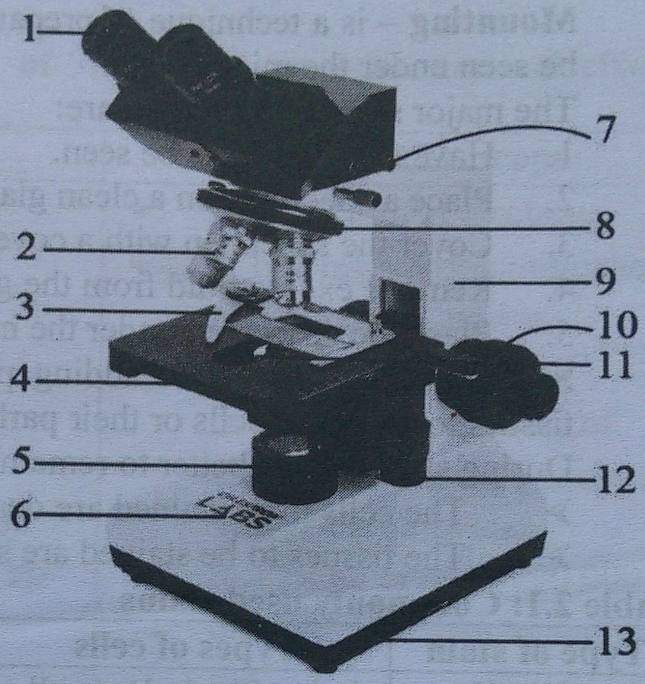


**Fig. 2.1. A compound Microscope**

## Parts of a compound Microscope

A compound microscope has the following major parts:

**Base** – supports the weight of the microscope.

**Mirror** – reflects light upward through the diaphragm and hole in the stage.

**Fine adjustment** – permits exact focusing by moving stage or body tube up or down very slightly.

**Diaphragm** – controls the amount of light passing through the specimen.

**Objectives** – contain lenses of different magnification.

**Stage** – supports the slide.

**Nose piece** – can be rotated to use the desired objective lens.

**Arm** – supports body tube and course adjustment.

**Course adjustment** – moves the body tube up and down.

**Body tube** – holds the nose piece at lower end and ocular lens at upper end.

**Ocular (eye piece) lens** – contains the lens to increase magnification

## Using the light Microscope

The basic techniques of using the light microscope include importa procedures like **mounting** and **staining.**

i) **Mounting** – is a technique of preparing an object or specimen be seen under the microscope.
The major steps of mounting are:
1. Have a specimen to be seen.
2. Place a specimen on a clean glass slide.
3. Cover the specimen with a cover slip.
4. Remove excess fluid from the glass slide.
5. Place the spacemen under the microscope.

ii) **Staining**: is a method of adding pigmented chemicals to sli tissues to make the cells or their parts easier to see.
During staining it is better to remember that:
- The cells to be stained are dead.
- The tissues to be stained are sliced very thinly.

**Table 2.1: Commonly used stains**

| Type of stain | Types of cells | Main organelles staine |
|---|---|---|
| Hematoxylin | Animal and plant cell | Nuclei stained blue/ pur or brown |
| Methylene blue | Animal cells | Nuclei stained blue |
| Acetocarmine | Animal and plant cell | Staining chromosomes dividing nuclei |
| Iodine | Plant cells | any material containing starch |

## The electron Microscope

For a long time, even after the formulation of cell theory only little w known about the fine details of the cells until the invention of **electro microscopes.**
The first electron microscope was developed in 1930s and came into us in 1950s. Thus, the discovery of electron microscope played a huge ro in increasing our biological knowledge. The electron microscope ca resolve an object at much smaller wavelengths.



The image that is directly. The electro which are focused by

**Table 2.2: Com Mic**

| Light Mic |
|---|
| • It uses beam<br>• It has relativ magnificatio resolution po<br>• The highest capacity of t microscope times and re to 500 time human eye. |
| **Advantages of microscope** |
| • It helps us cells.<br>• It is relati<br>• Easy to m to place.<br>• Do not re source. |
| **Disadvantage microscope** |
| • It has l and reso |

The image that is formed by electron microscope cannot be seen directly. The electron beam itself consists of the charged electrons which are focused by glass lenses in a light microscope.

**Table 2.2: Comparison of light Microscope and electron Microscope**

| Light Microscope | Electron Microscope |
|---|---|
| • It uses beam of light<br>• It has relatively less magnification and resolution power<br>• The highest magnification capacity of the light microscope is up to 2000 times and resolution of up to 500 times than the human eye. | • It uses beam of electrons<br>• It has highest magnification and resolution power.<br>• The maximum magnification capacity of electron microscope is up to 2,000,000 times and resolution of up to 200,000 times greater than human eye. |
| **Advantages of light microscope** | **Advantages of electron microscope** |
| • It helps us to study living cells.<br>• It is relatively cheap.<br>• Easy to move from place to place.<br>• Do not relay on electric source. | • It has high magnification and resolution power.<br>• It helps up to see three dimensional picture of surface of the specimen. |
| **Disadvantages of light microscope** | **Disadvantages of electron microscope** |
| • It has less magnification and resolution power | • It cannot help us to study the living cells<br>• It is very expensive<br>• It relies on electric source<br>• It is not easy to move from place to place. |

## Illustrative Example

1. The lenses of a compound microscope have the following magnification power. The eye piece 8×, low power objective 4×, middle power of objective 10× and high power objective 43×. If you observe your specimen under the middle power objective, how much would it be magnified?

   A. 80× B. 32× C. 344× D. 40×

   ***Explanation:*** Total magnification power of a compound microscope is the product of the eye piece and objective lens used. Accordingly the eye piece is 8× and middle power objective lens used is 10×. $8 \times 10 = 80\times$ **Answer: A**

2. What is the resolution of a microscope? It is the,

   A. scattering of the parts of the object to show more details
   B. Brightening of the parts of the object to show more details
   C. coloring of small objects to make them visible
   D. magnifying of small objects to make them visible

   ***Explanation:***
   A microscope has two abilities. These are magnification and resolution. Magnification means multiplying or increasing the size of an object. Resolution means scattering the detail or parts of an object. **Answer: A**

3. A microscope with two lens systems is called:

   A. compound microscope
   B. simple microscope
   C. Ocular microscope
   D. high power microscopes

   ***Explanation:***
   Compound microscopes have two lens systems. Namely, eye piece (ocular lens) and objective lenses (low power, middle power, high power and oil immersion lenses). **Answer A**

**Item number 4 is based on the following list of procedures which shows mounting a specimen for study under a microscope.**

1. Remove excess fluid from the glass slide.
2. Cover the specimen with a cover slip
3. Place the specimen on the glass slide.
4. Place the specimen under the microscope.

4. The correct order in mounting the specimen is:

   A. $2 \longrightarrow 3 \longrightarrow 1 \longrightarrow 4$ C. $3 \longrightarrow 2 \longrightarrow 1 \longrightarrow 4$
   B. $1 \longrightarrow 3 \longrightarrow 2 \longrightarrow 4$ D. $1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4$

***Explanation:*** Mounting is preparation of an object to be seen under the microscope. The steps of mounting include :

1. Place the specimen on the glass slide
2. Cover the specimen with a cover slip.
3. Remove excess fluid from the glass slide
4. Place the specimen under the microscope

**Answer: C**

5. Which of the following is the advantage of light microscope? (EGSECE, 2005 E.C)
   A. It can be used without electricity
   B. It shows a limited detail
   C. It can be kept in a separate room
   D. It has limited resolving power.

***Explanation:***

| **Advantages of light microscope** | **Advantages of electron microscope** |
|---|---|
| • Can be used without electricity<br>• Relatively less expensive<br>• Easy to move from place to place<br>• Can help to see a living cell | • It has highest magnification and resolution power.<br>• Helps to see 3 – D vision |

**Answer: A**

6. Suppose you are observing a specimen under the microscope which is magnified 600x the actual size of the specimen. If the magnification power of the eye piece lens is 15x, what is the magnification of the objective lens?
   A. 600x B. 40x C. 30x D. 60x

***Explanation:***

$$\text{The magnification of objective lens} = \frac{\text{Total magnification}}{\text{magnification of eyepice}}$$

$$= \frac{600}{15} = 40x$$

**Answer: B**

7. This item is based on the following list of steps in using a microscope. (EGSECE, 2006 E.C)
   1. Switch on the built in lamp.
   2. Clip the prepared slide.
   3. Set up the microscope with lowest power lens.

4. Look through the eye piece lens by turning the course focusing knob.
5. Adjust the first diaphragm.

The correct order of using a microscope is:

A. $3 \longrightarrow 1 \longrightarrow 2 \longrightarrow 4 \longrightarrow 5$
B. $3 \longrightarrow 4 \longrightarrow 5 \longrightarrow 2 \longrightarrow 1$
C. $3 \longrightarrow 2 \longrightarrow 1 \longrightarrow 5 \longrightarrow 4$
D. $3 \longrightarrow 5 \longrightarrow 2 \longrightarrow 1 \longrightarrow 4$

***Explanation***: The correct order during using microscope is:

1. Set up the microscope with lowest power lens.
2. Clip the prepared slide.
3. Switch the built lamp.
4. Adjust the first diaphragm.
5. Look through the eye piece lens by turning the course.

**Answer: C**

8. What is the resolution power of a microscope? It is the ability to: (EGSECE, 2006 E.C)

A. Show objects clearly.
B. Increase the size of objects
C. Make objects look smaller.
D. Make objects look brighter.

***Explanation***: A microscope has two major abilities. These are **resolution** and **magnification.**

- ***Resolution*** – ability to show the detailed or scattered parts of an object in order to see a clear image of an object.
- ***Magnification*** – increasing the size of an object

**Answer: A**

9. A grade 9 student wants to study onion epidermal cells using light microscope. Which of the following activities is correct during his observation? He will (EGSECE, 2006 E.C)

A. Stain the cells using methylene blue that reacts and turns the nucleus blue.
B. Focus the lenses to see ribosome and mitochondria within the cells.
C. Stain the cells using iodine that reacts with the starch of the cells and turns blue black.
D. Start to set up the microscope with the highest power lens in place.

***Explanation:***

- Among the most commonly used stains. The most widely used ones are:
  - **Mnethylene blue** – stains the nucleus of animal cells blue.
  - **Iodine** – turns any plant material containing starch blue black
  - **Acetocarmine** – for staining chromosomes in dividing nuclei.
  - **Hematoxylin** – stains the nucleus of animal and plant cell blue/ purple/ brown.

**Answer: D**

10. If you are given a compound light microscope with the following magnification of its objectives.
Low power objective = 15×
Middle power objective = 30×
High power objective = 75×
Oil immersion objective = 150×

Then, calculate the total magnification of the microscope under the middle power objective, if the magnification power of the eye piece is one fifth of that of the high power objective lens system show your work clearly.

***Explanation:***

Eyepiece lens = $\frac{1}{5}$ of high power objective lens

$= \frac{1}{5} \times 75 = 15\times$

Total magnification = eyepiece × middle power objective power
$= 15 \times 30 = \underline{450\times}$

11. If 0.01mm size sample is observed by a microscope having a magnification of 7× and 10 × of its ocular lens and middle power objective lens respectively. What will be the size of the sample after magnification?

***Explanation:***

$= 0.01\text{mm} \times 7 \times 10 = \underline{0.7\text{mm}}$

12. How many lens system do a compound microscopes have?
A. Three B. Four C. One D. Two

***Explanation***: A compound microscope is a microscope which has two lens systems. These are **eyepiece (ocular) lens** and **objective lens system.**

- Objective lenses can be:
  - Low power
  - Middle power
  - High power
  - Oil immersion lenses
- The total magnification power of a compound microscope is the product of magnification of eye piece and objective lens.

**Answer: D**

13. Which of the following is important for a better contrast in microscopic observations?
    A. Adding alcohol to specimen
    B. Covering specimen with cover slip
    C. Staining specimen
    D. Avoiding water to specimen

    ***Explanation***: Stains are chemicals which are added to the specimen in order to see the clearly refined and colorful structures of objects to be seen.

    **Answer: C**

14. Describe the advantages and disadvantages of light microscope and electron microscopes.

    ***Explanation***: see table 2.2

15. Explain the meanings of:
    - Magnification
    - Staining
    - Mounting
    - Resolution

    ***Explanation***:
    - ***Magnification***: ability of the microscope to increase the size of an object.
    - ***Resolution***: ability of the microscope to show the detailed and scattered parts of an object.
    - ***Staining***: use of chemicals **(stains)** to see a bright and colorful picture of a specimen and its parts.
    - ***Mounting***: preparation of a specimen to see under the microscope.

16. List out some types of stains which are used in biological laboratories.

    ***Explanation***: see table 2.1.



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***Part I:*** **Choo**

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   A.

3. The
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   A.
   B.
   C.
   D.

17. What are the contributions that electron microscope play in increasing our biological knowledge?
*Explanation*: Electron microscope has the highest magnification and resolution power. Thus, electron microscope helped biologists to know more about cells and tissues of an organism. Even more, it helped the microbiologists to know more about microorganisms.

18. List out some of the disadvantages of electron microscope over light microscope.
*Explanation*: Even though, the electron microscope has highest resolving and magnification power, it has the following disadvantages:
- It is very expensive.
- It is difficult to move from place to place
- It relies on electric source.
- It do not enable us to study living cells.

19. Why do we need microscope to see a unicellular organism?
*Explanation*: Unicellular organisms are very small and we cannot see them by using our naked eye. Thus, microscope is a tool which has highest magnifying and resolving capacity than our eye. Accordingly, we can only see and study unicellular organisms by using a microscope.

## *Post Test I*

***Part I:* Choose the best answer from the given alternatives.**

1. Which of the following tool of a biologist is used to see very small objects that cannot be by our naked eyes?
   A. Telescope C. Stethoscope
   B. Microscope D. Thermometer

2. If an object having a size of 0.01 mm is seen by using a microscope having an eye piece of 10X and objective piece of 10X, what is the magnified size of an object in mm?
   A. 1mm B. 0.1mm C. 10mm D. 0.01mm

3. The disadvantage of using light microscope as compared to electron microscope :
   A. Is relatively cheap
   B. easy to move from place to place
   C. do not relay on electric source
   D. has less magnification and resolution power

4. The ability of a microscope to show the detailed structure of an object is:
   A. Magnification
   B. Resolution
   C. Staining
   D. mounting
5. The part of a microscope that reflects light upward through the diaphragm and hole in the stage is:
   A. Mirror B. stage C. base D. Diaphragm
6. The total magnification size of a compound microscope is:
   A. The sum of objective and ocular lens
   B. The product of objective and ocular lens
   C. The average of objective and ocular lens
   D. The difference of objective and ocular lens
7. A stain that is used to see any material containing starch is:
   A. Hematoxylin
   B. Methylene blue
   C. Iodine
   D. Acetocarmine
8. A technique of preparing an object or specimen to be seen under the microscope is:
   A. Staining
   B. Mounting
   C. layering
   D. Pressing
9. Which of the following is true about the resolving power of a microscope?
   A. The higher the resolving power, the lower the magnification
   B. The higher the resolving power, the more detail it can show
   C. The higher the resolving power, the less detail it can show
   D. The lesser the resolving power, the more detail it can show
10. In mounting a specimen, which stage comes last?
   A. Place the specimen on the glass slide
   B. Cover the specimen with a cover slip.
   C. Remove excess fluid from the glass slide
   D. Place the specimen under the microscope

## 2.2. The cell

All living organisms are made up cells. Although some living systems consist of only one cell and others of several billion even most complex organisms begin life as a single cell, fertilized egg. Like the bricks of a building, cells are the building blocks of the organism

Some organisms like bacteria and protozoa are composed of single cell and are called **unicellular organisms**. While others like plants, animals, most fungi and algae are composed of many cells and are called **multicellular organisms.**

Therefore, all living organisms share certain characteristics. The seven characteristics that all living organisms share in common are:

- **Nutrition** – is the process by which organisms obtain and utilize food to provide energy for the body some organisms can synthesize their own food and are called **authotrophs**. Others cannot synthesize their own food and are called **heterotrophs.**
- **Respiration** – the process by which living organisms get energy from the food.
- **Excretion** – the process of getting rid of metabolic wastes produced by the cell.
- **Growth** – is increase in size and mass.
- **Irritability** – sensitivity to changes in the surrounding environment.
  - **Stimuli** – an environmental change that makes an organism to respond.
  - **Response** – the reaction of an organism to the stimuli.
- **Movement** – the need to get near or away from things. Animals move using their muscles, plants move through growth.
- **Reproduction** – the production of offspring to ensure the survival of the species. It is very important to keep continuity of life.

## Cell theory

Cells were first discovered by English scientist Called **Robert Hooke** in 1665. By using a microscope he made by his own he examined a thin slice from the cork.

He finally observed tiny empty compartments in the cork and called them **cellule** a Latin word meaning '**little rooms**'. This term came down to us as cells. However, the first cells seen by Robert Hook were dead cells.

The first living cells were observed by a Dutch naturalist called **Anton Van Leeuwenhoek**. He called tiny organisms which he observed "**animalcules**" meaning **little animals**.

In 1838 after careful study on plant tissues, the German botanist called **Mathias Schielden** stated that 'All plant tissues are composed of cells.'

A year after, another German Zoologist **Theodore Schwann** reported that 'all animal tissues are composed of cells.'

Therefore, the cell theory consists of the following major points:

1. All living organisms are composed of one or more cells.
   - ✓ ***Unicellular organisms*** are composed of a single cell.
     **Example**: Bacteria, protozoa, yeast, etc
   - ✓ ***Multicellular organisms*** are composed of many cells.
     **Example:** Animals, plants, etc.
2. Cells arise only by division from previously existing cell.
3. Cells are the basic structural and functional units of life.

## Eukaryotic and Prokaryotic cells

Organisms can be grouped in to two fundamental types based on the structure and complexity of their cells.

### i) Prokaryotic cells

In prokaryotic cells the plasma membrane confines the contents of the cell to an internal compartment, but they do not have distinct internal membrane systems in the form of organelles.

The nucleus and other organelles are not bounded by double membrane system. Therefore, the nuclear materials freely suspend in the cytoplasm and some organelles found in other cells might be missing.

**Examples** for prokaryotic organisms are **bacteria** and **blue green algae.**

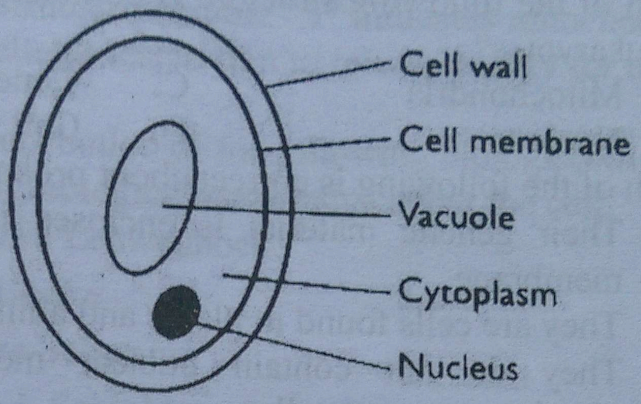


**Fig. 2.2: The Structure of bacterial cell**

**ii) Eukaryotic Cells**

Eukaryotic cells are cells which posses' membrane bounded structures. Nucleus and organelles are double membrane bounded. These types of cells are found in all organisms except bacteria and blue green algae.

**Illustrative Example**

20. Which of the following is not true of the cell theory?
    A. All organisms are composed of cells.
    B. Cells arose spontaneously from organic compounds.
    C. Cells are the structural and functional units of life.
    D. Cells are the basic units of living things.

    *Explanation*: The major points of cell theory include:
    - Organisms are composed of one or more cells.
    - Cells are the basic structural and functional units of life
    - Cells arise only by division of previously existing cell.

    **Answer: B**

21. If you observe a cell with its nucleus and other organelles lacking membranes, the cell could be:
    A. Bacterium C. an egg
    B. A neuron D. an amoeba

    *Explanation*: prokaryotic cells lack membrane bounded organelles and nucleus. Examples for these types of cells are bacterium and blue green algae. **Answer: A**

22. Which of the following structure is common to both prokaryotes and eukaryotes?
    A. Nuclear membrane C. Chromatin materials
    B. Mitochondrial membrane D. Cell wall

    **Answer: C**

23. Which of the following structure is common to both prokaryotes and eukaryotes?
A. Mitochondria
B. Nucleus
C. Genetic material
D. Cell wall
**Answer:** C

24. Which of the following is correct about prokaryotic cells?
A. Their genetic material is enclosed in a double nuclear membrane
B. They are cells found in plants and animals.
C. They do not contain nuclear membrane and other membranous organelles.
D. such cells are found in fungi and protozoan's.

***Explanation:*** Comparison of prokaryotic and eukaryotic cells

| **Prokaryotic cells** | **Eukaryotic cells** |
|---|---|
| • The nucleus and other organelles are not bounded by membrane.<br>• The nuclear materials such as DNA and chromosomes freely suspend in the cytoplasm<br>• Some organelles are missing.<br>***Example***: Bacteria and blue green algae | • The nucleus and other organelles are double membrane bounded.<br>• The nuclear materials are enclosed by the nuclear membrane.<br>• Contain full structures and organelles.<br>***Example***: All organisms except bacteria and blue green algae |

**Answer:**

Question number 20 and 21 are based on the following illustration of the cell. (EGSECE 1995 E.C)

25. Which structure immediately identifies this cell as eukaryote?
A. Structure "3"
B. Structure "4"
C. Structure "1"
D. Structure "2"

***Explanation***: The cell indicated above is animal cell. Animal cells are eukaryotic cells in which they have true nucleus which is bounded by membrane. In addition, eukaryotic cells have double membrane bounded organelles. Nucleus is indicated by number '2' on the above figure.
**Answer:** D

26. The cell uses structure "3"
A. As a receptor
B. To produce energy
C. To transport material from one part of the cell to the other
D. To package proteins so they can be stored by the cell.

***Explanation***: Structure number "3" indicates mitochondria which is the site of cellular respiration to release energy for the cell.

**Answer: B**

27. Discuss the contribution of the following biologists in the development of biological knowledge about the cell.
    - Anton Van Leuwenhoek
    - Robert Hooke
    - Theodore Schwann
    - Mathias J. Schielden

***Explanation***:

Even though various biologists have made a great contribution about our biological knowledge of cell, some of the major ones are:

- **Anton Van Leeuwenhoek** – invented a microscope and first person to see animal cells.
- **Robert Hooke** – the first person to discover cells,
- **Theodore Schwann** and **M.J. Schielden** formulated the cell theory.

28. The statement of cell theory which disproves the theory of spontaneous generation is:

***Explanation***:

The theory of spontaneous generation states that 'certain non living things can be directly changed in to a living thing.' However, the cell theory states that, "All cells came from pre – existing cell: Thus, a cell can never come from a non living thing, instead one cell comes from another cell.

29. List out the distinguishing Features of prokaryotes from eukaryotic organisms.

***Explanation***:

Prokaryotic organisms like bacteria and blue green algae differ from eukaryotic organisms by:

- They have no true nucleus in which nuclear materials are bounded by nuclear membrane.
- They do not have double membrane bounded organelles.
- They lack some structures that are present in eukaryotic cells.

30. This item is based on the table below which shows the different life processes and their accomplishment.

| No. | Life process | Accomplishment |
|---|---|---|
| 1 | Excretion | Getting rid of the waste products of digestion |
| 2 | Reproduction | Producing offspring |
| 3 | Growth | Increase both in size and mass |
| 4 | Irritability | Being sensitive to changes in the environment |

Which life process is NOT correctly matched with its accomplishment?

A. 3 B. 4 C. 1 D. 2

*Explanation*:

- The common life process of living things are:
  1. **Excretion** – the process of getting rid of metabolic wastes.
  2. **Reproduction** – process of producing new off springs.
  3. **Growth** – is increase in size and mass.
  4. **Irritability** – is sensitivity of an organism to its environment.
  5. **Nutrition** the process of obtaining and utilizing food
  6. Respiration – The process by which organisms obtain energy from food.
  7. **Movement** – need to get near to or away from things.

**Note:** The process of removing wastes of digestive products is called ***egestion (defecation).***

- Excretory products are produced as a result of metabolic processes in the cells. **Answer:** C

## *Post Test 2*

***Part I: Choose the best option from the given alternatives.***

1. Which one of the following is prokaryote?

A. Protozoa
B. Algae
C. Bacteria
D. plant

2. Which one is common to prokaryotes and eukaryotes?
   A. Nuclear materials
   B. Membrane bounded organelles
   C. Nuclear membrane
   D. Cell wall
3. All are the common characteristics of living things, except:
   A. Reproduction
   B. Photosynthesis
   C. Growth
   D. Nutrition
4. Cells were first discovered by:
   A. Anton Van Leuwenhoek
   B. Louis Pasteur
   C. Robert Hooke
   D. Robert Koch
5. One of the following is not a point of cell theory:
   A. Organisms are composed of one or more cells
   B. Cells can arise spontaneously from non living things
   C. Cells come from pre-existing cells
   D. Cells are structural and functional units of living things

***Part II: Matching***

| ***Column A*** | ***Column B*** |
|---|---|
| 1. Reproduction | A. Increase in size and mass of the body |
| 2. Nutrition | B. Feeding |
| 3. Sensitivity | C. Getting rid of metabolic wastes |
| 4. Movement | D. Release of energy from the food |
| 5. Excretion | E. production of offspring to ensure the survival of the species |
| 6. Respiration | F. the need to get near or away from things |
| 7. Growth | G. Response to change |

## Eukaryotic cell structures and their functions

Eukaryotic cells have three major parts. These are; **nucleus**, **cytoplasm** and **cell membrane**.

- **Nucleus** – controls and directs all the activities of the cell. It contains the DNA (Deoxyribonucleic acid) which makes up the **Chromosomes**.

*Note*: **Chromosomes** – are the genetic materials which contain instructions for making an organism.

- **Cell membrane** - is semi/partially permeable membrane that controls the movement of substances in and out of the cell. Some of the substances that pass across the cell membrane are **carbondioxide,** oxygen and water. It is made up of protein and lipid bi layer. Thus, it is called **a lipoprotein layer.**
  The cell membrane is a living layer of a cell which is found on the external layer of animal cells, while in plant cells it is found next to the cell wall.
- **Cytoplasm** – is the whole internal structure of the cell without the nucleus. It is filled with jelly substance and consists of thousands of structures called **organelles.**
  The cytoplasm is a center for the chemical reactions of the cell. The major organelles that are found in the cytoplasm are; **mitochondria, lysosomes, vacuoles, plastids, centrioles, Golgi complex, endoplasmic reticulum, etc.**

**Note**: The word **organelle** is derived from a Latin word Meaning **'little organs'**

1) **Mitochondria** (*Singular*; Mitochondrion) – are called the power house of the cells since they are sites of most of the reactions of cellular respiration. Thus they are responsible for releasing energy for the cellular activities.
   Mitochondria are surrounded by two membranes. The inner membrane is folded in to the structure called **cristae**. The folds help to increase its surface area for the respiratory reactions. The outer membrane is smooth and it allows small molecules to pass through it.
2) **Endoplasmic reticulum** – is the network of internal membranes extending through the cytoplasm. It connects the nucleus with the cell membrane. There are two forms endoplasmic reticulum in a cell. These are:
   - **Rough Endoplasmic reticulum** – have ribosome on its membrane surface and is involved in protein transport which is synthesized by ribosome.
   - **Smooth Endoplasmic reticulum** – have no ribosome on its surface. It is concerned with lipid synthesis, carbohydrate metabolism and detoxification.

3) **Ribosome** – are found attached to rough endoplasmic reticulum and some are free in the cytoplasm. They are composed of RNA and protein.
   - Ribosomes are involved in protein synthesis.

4) **Golgi complex (Golgi body)** – are sacks of flattened membrane which modifies proteins, packages and concentrates materials for transport. This organelle was first described by an Italian microscopist **Camilo Golgi.**

**Some of the organelles found in animal Cells but not in plant cells**

- **Lysosomes** are membranous sacks that produce digestive enzymes and disintegrates damaged and worn out cells. Thus, they are sometimes called ***'self destructing organelles.'***
- **Centrioles** are pair of hallow cylinders located near the center of the cell. They form mitotic spindle fibers during animal cell division.

**Some of the structures and organelles Found in plant cell but not in animal cell**

- **Central vacuole** is a fluid filled sack which contains a fluid called **cell sap.** They are used to transport and store ingested materials, wastes and water. They are also used to maintain **turgidity** of cell.
- **Plastid** consists of three structures. These are **chloroplast, chromoplast** and **leucoplast**.
  i) ***Chloroplast*** – are sites of photosynthesis. They also contain a light absorbing pigment called **chlorophyll.**
  ii) ***Chromoplast*** – gives color to flowers and fruits.
  iii) ***Leucoplast*** – storage site of starch.
- **Cell wall** – is the outer layer of plant cells, which is mainly composed of complex carbohydrates called **cellulose.** It provides support, shape and protection for a cell. The cell wall is a completely permeable membrane.

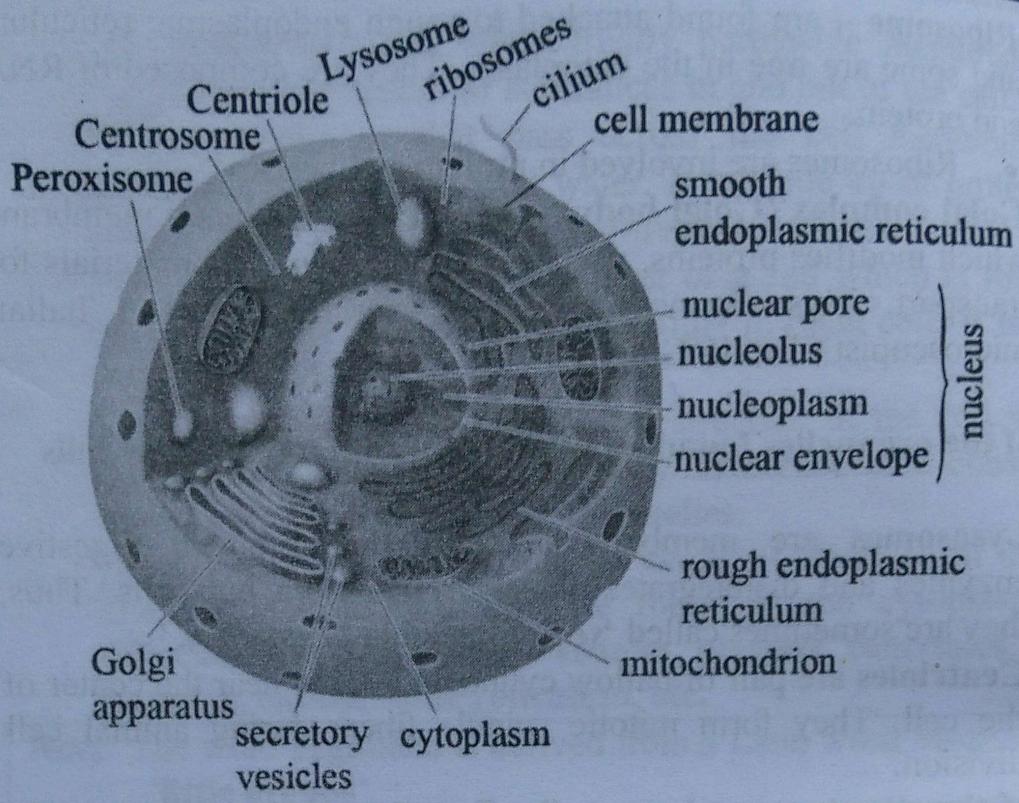


**Fig. 2.3: Generalized animal cell**

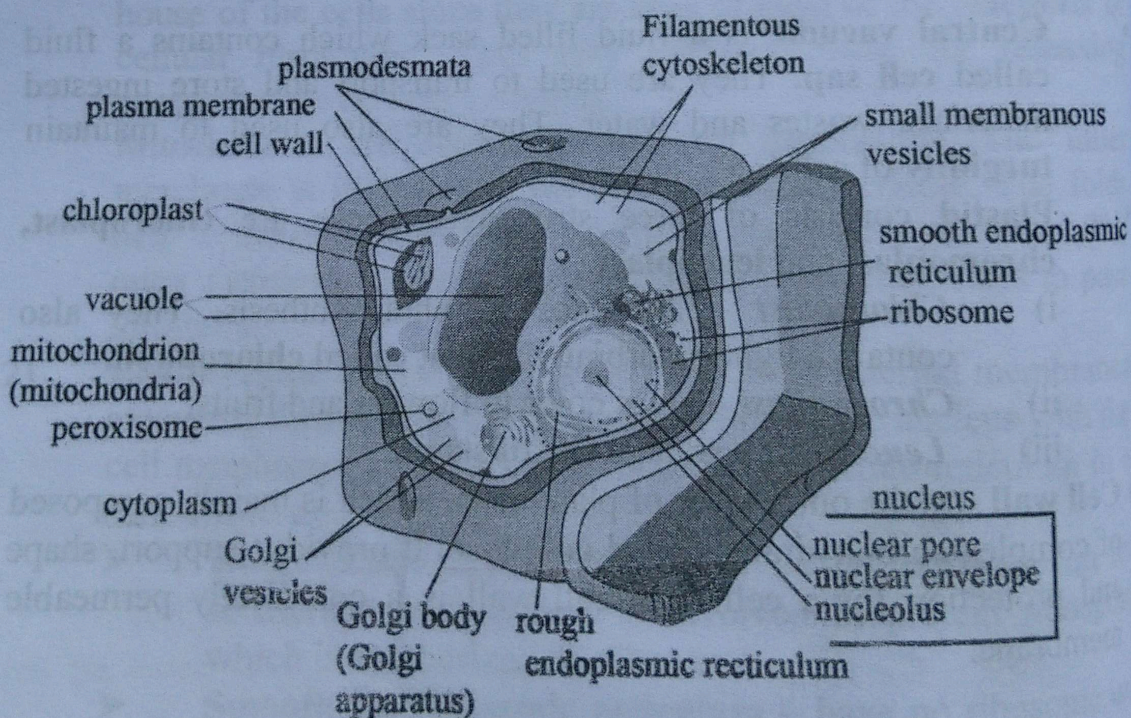


**Fig. 2.4 Generalized plant cell**

## Illustrative Example

31. Cells that are adapted to carry out a particular body function are:
    A. undifferentiated cells
    B. embryonic stem cells
    C. Unspecialized cells
    D. specialized cells

*Explanation:*
- **Unspecialized cells** – are cells that have not yet assumed to perform their final functional characteristics.
- **Embryonic stem cells** – are cells from the early embryo that have the potential to form almost any other type of cell.
- **Differentiated cells/ specialized cells** – are special cells which carry out specific functions.

**Answer: D**

32. All living cells can be characterized by one of the following.
    A. All process cytoplasm and nuclear materials.
    B. They posses nucleus surrounded by nuclear membrane.
    C. Are capable of sexual multiplication.
    D. They posses cell wall.

*Explanation*: All living cells, both prokaryotic and eukaryotic cells posses cytoplasm and nuclear materials.

**Answer: A**

33. Which of the following structures of a cell are involved in the synthesis and transportation of proteins?
    A. ribosome and centriole
    B. Lysosome and mitochondria
    C. Ribosome and rough endoplasmic reticulum
    D. Ribosome and lysosomes

*Explanation:*
- ✓ Ribosome is involved in protein synthesis.
- ✓ Rough Endoplasmic reticulum has ribosome on its surfaces and its used to transport proteins inside a cell.

**Answer: C**

34. Which of the following parts of the cell is made up of lipoprotein?
    A. Ribosome
    B. plasma membranes
    C. cell wall
    D. DNA

*Explanation:* Plasma membrane /cell membrane/ is made up protein and lipid bilayers.

**Answer: B**

35. The colors of flowers are due to the presence of pigments kno as:
A. Leucoplasts
B. lysosomes
C. chloroplasts
D. chromoplasts

*Explanation*: plastid is an organelle which is found only in plan cells. It has three parts:

- ***Chloroplast*** – consists of light absorbing and gre pigmented substance known as chlorophyll
  - It's the site of photosynthesis in plant cells.
- ***Chromoplast*** – gives color to fruits and flowers.
- ***Leucoplast*** – The storage site of starch in plant cells.

**Answer:**

36. The 'suicide bag' of a cell is:
A. centriole
B. lysosomes
C. mitochondria
D. vacuole

*Explanation*:

**Lysosome** – an organelle which is found only in animal cells.
- It helps to produce digestive enzymes and to break disintegrate damaged and worn out cells. **Answer: B**

37. A cell with large number of mitochondria is expected to
A. Produce more protein
B. make a lot of ATP
C. perform photosynthesis
D. store more food

*Explanation*:

- **Mitochondria** – is the main site of cellular respiration.
- **Respiration** – the process in which energy is release from the food.
- **ATP (Adenosine tri phosphate)** – the direct an immediate source of energy for a cell.

Therefore, a cell with large number of mitochondria (Example Muscle cell, liver cell) are expected to release more energy in th form of ATP. (For more detail see under the part of respiration i chapter 3)

38. Explain why do cells of the body like muscle cell and liver cel posses large number of mitochondria.

**Answer:** B

*Explanation*: Since mitochondria is the site of cellular respiration where energy is released for cellular activities, cells like muscle cell, liver cell and brain cell that require more energy do have large number of mitochondria than any other cells of the body.



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**Part I: Ch**

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   B.
7. A di
   A
   B

39. An organelle that is important for protein synthesis is known as: (EGSECE, 2006 E.C)
A. Chloroplast
B. Ribosome
C. Mitochondrion
D. nucleus

*Explanation:*

| Organelle | Function |
|---|---|
| Chloroplast | The site of photosynthesis |
| Ribosome | Involved in protein synthesis |
| Mitochondria | The site of cellular respiration |
| Nucleus | Controls and directs the cellular activities |

**Answer: B**

## *Post Test 3*

**Part I: Choose the best option from the given alternatives.**

1. Chromosomes that contain the genetic information of an organism are found in the:
A. Cell wall
B. Nucleus
C. Cell membrane
D. Chloroplast

2. Energy releasing organelle of a cell is:
A. Ribosome
B. Centriole
C. Mitochondria
D. Lysosome

3. An organelle of cell that synthesizes protein is:
A. ribosome
B. Golgi body
C. mitochondria
D. nucleus

4. Which one of the following includes all the rest?
A. Nucleolus
B. Nucleus
C. Cytoplasm
D. Protoplasm

5. All of the following are included in the cytoplasm except:
A. mitochondria
B. ribosome
C. nucleus
D. Golgi body

6. The structure that keeps plant cells turgid is:
A. Vacuole
B. Cell wall
C. nucleus
D. ribosome

7. A cellular structure in animals that assists the process of cell division is:
A. lysosome
B. centriole
C. nucleus
D. cell membrane

8. All are the major structures of a cell except one. Which one is it?
   A. Cell membrane
   B. Cell wall
   C. Cytoplasm
   D. Nucleus
9. The movement of substances that move in and out of a cell controlled by:
   A. cell wall
   B. nucleus
   C. endoplasmic reticulum
   D. cell membrane
10. Cellular activities are controlled and directed by:
   A. Cell membrane
   B. Cytoplasm
   C. mitochondria
   D. Nucleus

## Comparing plant and animal cells

### Plant cells

Consists of an outer layer called cell wall.

- **Cell wall** – is composed of complex organic materia called cellulose.
  - It provides support, shape and protection for a cell.
- It has regular shape.
- Some organelles that are found plant cell but not in animal cell include; plastid (chloroplast, chromoplast, leucoplast), large central vacuole, etc.
- Have larger size than animal cells.

### Animal cells

- Cell membrane is an outer layer and has no cell wall.
- It is irregularly shaped.
- Vacuole is small and temporary.
- Organelles found in animal cell but not in plant cell are lysosome and centriole.

➢ Structures that are commonly found in plant cell and animal cell are **cell membrane, mitochondria, endoplasmic reticulum, Ribosome, Golgi complex, nucleus,** etc.



***Table:*** **Compariso**

| Plant |
| --- |
| • Have regular s<br>• They are large<br>• They have cell<br>• They have or plastid, centra<br>• The vacoule permanent. |

40. Which of plant and
   A. Ch
   B. Ce
   C. M
   D. C

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| *Structures common to and animal c* |
| --- |
| Mitochondria |
| Cell membra |
| Endoplasmic |
| Ribosome, |
| Golgi body |

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**plant cell and animal cell**

| Plant cell | Animal cell |
|---|---|
| • Have regular shape.<br>• They are larger in size.<br>• They have cellulose cell wall<br>• They have organelles such as plastid, central vacuole.<br>• The vacoule is central and permanent. | • Have irregular shape.<br>• They are relatively small in size.<br>• They have no cell wall.<br>• They have organelles such as centriole and lysosomes.<br>• The nucleus is central.<br>• The vacuole is temporary. |

## Illustrative Examples

40. Which of the following are distinguishing characteristics between plant and animal cells?
    A. Chloroplast and mitochondria
    B. Cell wall and cell membrane
    C. Mitochondria and nucleus
    D. Cell wall and chloroplast

***Explanation*: Comparison of plant and animal cells**

| *Structures that are common to both plant and animal cell* | *Structures found only in plant cell* | *Structures found only in animal cell* |
|---|---|---|
| Mitochondria, nucleus, Cell membrane , Endoplasmic reticulum, Ribosome, Golgi body | Cell wall, Chloroplast, Large and central vacuole | Centriol<br>Lysosome |

Therefore; plant cells differ from animal cell by having **cell wall, chloroplast, large central vacuole, large size, regular shape.**

**Answer: D**

41. The outer most covering of all animal cells is:
    A. Protoplasm
    B. Plasma membrane
    C. cell wall
    D. vacuole

***Explanation*: Cell membrane /Plasma membrane/** is an organelle which is found in both plant and animal cells to control the movement of substances in and out of the cell. Moreover, it is the most outer layer (cover) of animal cells. The most outer layer for a plant cell is **cell wall.**

**Answer: B**

*Table:* **Comparison of plant cell and animal cell**

| Plant cell | Animal cell |
|---|---|
| • Have regular shape.<br>• They are larger in size.<br>• They have cellulose cell wall<br>• They have organelles such as plastid, central vacuole.<br>• The vacoule is central and permanent. | • Have irregular shape.<br>• They are relatively small in size.<br>• They have no cell wall.<br>• They have organelles such as centriole and lysosomes.<br>• The nucleus is central.<br>• The vacuole is temporary. |

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|---|---|---|
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**Answer: B**

42. The cell wall of plant cells is mainly made up of
A. Phospholipids
B. Cellulose
C. disaccharides
D. glycoprotein's

***Explanation***: Plant cell wall is made up of mainly a complex polysaccharide substance called **cellulose.**

➢ Plant cell wall provide support, shape and protection for a cell

**Answer: B**

43. Which of the following includes all the rest?
A. Cytoplasm
B. Protoplasm
C. cell vacuole
D. Nucleus

***Explanation:***

- A cell has three major parts. These are, cell membrane, cytoplasm and nucleus.
  - ➢ **Cell membrane** – is the outer and selectively permeable layer of a cell.
  - ➢ **Cytoplasm** – is the inside of the cell which is filled with jelly fluid and organelles. It excludes the nucleus.
  - ➢ **Nucleus** – is the part of the cell that controls and directs cellular activities. It also consists of genetic materials like chromosomes, DNA and RNA.

**Note: Protoplasm** – is the living part of a cell consisting of cytoplasmic structures and nucleus.

**Answer: B**

44. Both plant and animal cells contain
A. Cell wall, cell membrane, nucleus and cytoplasm
B. Cell membrane, cytoplasm, nucleus and mitochondria
C. Cell membrane, cytoplasm, nucleus and chloroplasts
D. Cell membrane, cytoplasm, nucleus, chloroplast and mitochondria

***Explanation***: The cellular structures which are commonly found in both plant and animal cells are:

- Cell membrane
- Cytoplasm
- Nucleus
- Mitochondria
- Endoplasmic reticulum
- Ribosome
- Golgi complex

**Answer: B**

45. The Golgi apparatus is involved in:
A. Altering or modifying proteins
B. Packaging proteins in to vesicles
C. Producing lysosomes
D. All of the above

**Answer: B**



46.

47.

48.

49.

50.

46. If you scarp the inside of your cheek, mount and examine it under a microscope, which of the following structures may not be seen?
A. Mitochondria C. cell membrane
B. Cell wall D. Nucleus
*Explanation*: Human cheek cells contain mitochondria, cell membrane, and nucleus. However, cell wall is not found in human cells.
- Cell wall is the outer most layer of cell of plants, algae, some fungi and bacteria.

**Answer: B**

47. Which of the following organelles is common to both plant and animal cells?
A. chloroplast C. Nucleus
B. cell wall D. vacuole
*Explanation*: Organelles that are commonly found in plant and animal cells are: ***cell membrane, nucleus, mitochondria, ribosome, endoplasmic reticulum, Golgi body***. **Answer: C**

48. Which of the following plant cell structures provides protection and support?
A. Vacuole C. cell membrane
B. cell wall D. Nucleus
*Explanation*: Cell wall is a structure which is found in plant cells, some bacteria and fungi. It is the most outer layer of these cells. Plant cell wall is made up of cellulose, while bacterial and fungal cell walls are made up of substances like murein. The main importance of cell wall in plant cells is, it provides support, protection and shape for a plant cell. **Answer: B**

49. Your teacher told you to mount an onion epidermal cell and list all structures you see under the microscope. Which of the following structures may not be included in your list?
A. plastids C. mitochondria
B. centrioles D. nucleus
*Explanation*: Centrioles and lysosomes are organelles that are found in animal cell but not plant cell. **Answer: B**

50. Which of the following structures occupies the central position in plant cells?
A. Nucleus C. Chloroplast
B. Vacuole D. Cytoplasm

***Explanation:*** Plant cells have large, permanent central vacuole While, animal cells have small and temporary vacuole.

**Answer:**

## *Post Test 4*

**Part I: Choose the best option from the given alternatives**

1. An organelle that is found in plant cell but not in animal cell is:
   A. Cell wall
   B. Mitochondria
   C. Lysosome
   D. Centriole
2. Which one of the following organelles is commonly found both in plant cell and animal cell?
   A. Centriole
   B. Cell membrane
   C. Lysosomes
   D. cell wall
3. Plant cells are different from animal cell because they:
   A. Have irregular shape
   B. Are relatively small
   C. Have central vacuole
   D. Are shapeless
4. Of the following pair of organelles which ones are found in animal cells but not in plant cells?
   A. Mitochondria and lysosome
   B. Centriole and Golgi body
   C. Golgi body and cell wall
   D. Centriole and lysosome
5. The support and protection of a plant cell is maintained by:
   A. Golgi body
   B. Central vacuole
   C. Ribosome
   D. Cell wall
6. Animal cells can photosynthesize but plant cells cannot because they have:
   A. Cell wall
   B. Nucleus
   C. Chloroplast
   D. Ribosome

***Part II*: Give short answer for the following.**

7. The following are list of organelles of a cell.
   - Cell wall
   - Cell membrane
   - Nucleus
   - Mitochondria
   - Golgi body
   - Central vacuole
   - Ribosome
   - Centriole
   - Lysosome
   - Endoplasmic reticulum
   - Chloroplast

Identify the ones which are found in:

- Plant cells only
- Animal cells only
- Both plant and animal cells

## 2.3. The cell and its environment

A cell interacts with its environment through the cell membrane. This is due to the reason that a cell requires useful nutrients and oxygen and also removes waste materials.

Exchange of materials between the cell and its environment takes place in the following major ways:

- Diffusion
- Osmosis
- Active transport

### i) Diffusion

**Diffusion** is the movement of molecules from the area of greater concentration to the area of lesser concentration until the molecules tend to distribute uniformly.

The rate of diffusion can be affected by factors like ***temperature*** and ***concentration***.

➢ As the temperature rises, the rate of diffusion also increases.

Diffusion occurs along **concentration gradient.**

**Concentration gradient** is the difference in concentration between two areas.

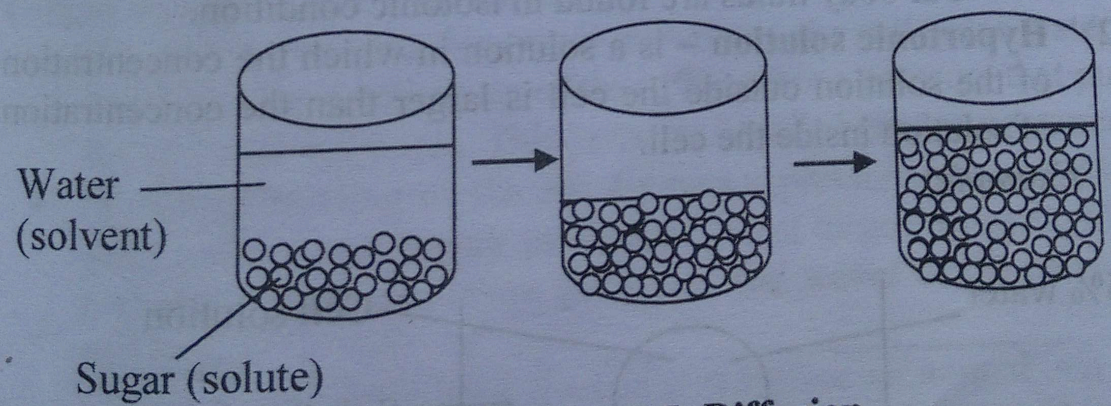


**Fig. 2.5: Diffusion**

***Osmosis*** is the movement of water molecules from an area of *higher concentration* to the area of *lower concentration* through a **semi permeable membrane**. It is also known as '***special type of diffusion.***'

***Semi- permeable membrane*** – which is also called selectively permeable is a membrane that allows some substances to pass through and stop others.

Thus diffusion is the movement of water molecules from dilute **solution (weak solution)** to **concentrated solution (strong solution).**

- ✓ ***Dilute solution (weak solution)*** – consists of high concentration of solvent and low concentration of solute.
- ✓ ***Concentrated solution (strong solution) is*** a solution that relatively consists of low solvent concentration and high concentration of solute.

Osmosis in living cells can be shown in the following ways.

1) **Isotonic solution** – is formed if solution of both sides of the cell (i.e. inside and outside the cell) are the same, there is no movement of water (solvent) on either sides.

***Example:***

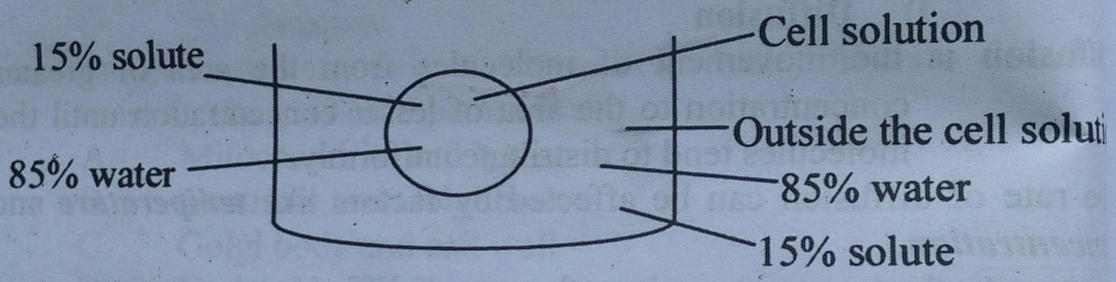


**Fig. 2.6: Isotonic Solution**

- Both solutions are the same; so there is no water movement across the cell membrane because of osmosis.
- Our body fluids are found in isotonic condition.

2) **Hypertonic solution** – is a solution in which the concentration of the solution outside the cell is larger than the concentration of solution inside the cell.

***Example:***

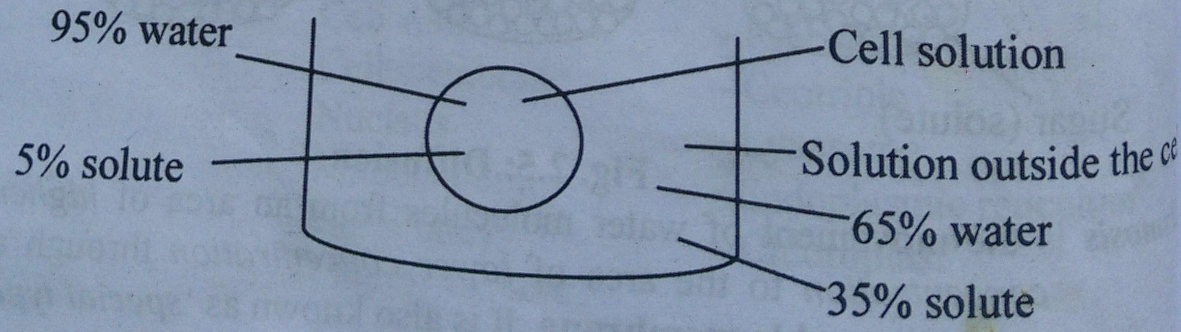


**Fig. 2.7: Hypertonic solution**

- Then because of osmosis, water molecules move from the region of higher concentration (the cell solution) to the region of lower concentration which is outside the cell (cell environment).
- In this condition water passes out of the cell or the cell loses its water content resulting in **shrinking** of the cytoplasm of the cell.

3) **Hypotonic solution** – is formed when the concentration of the solution outside the cell is lower than the concentration of the solution inside the cell.

***Example:***

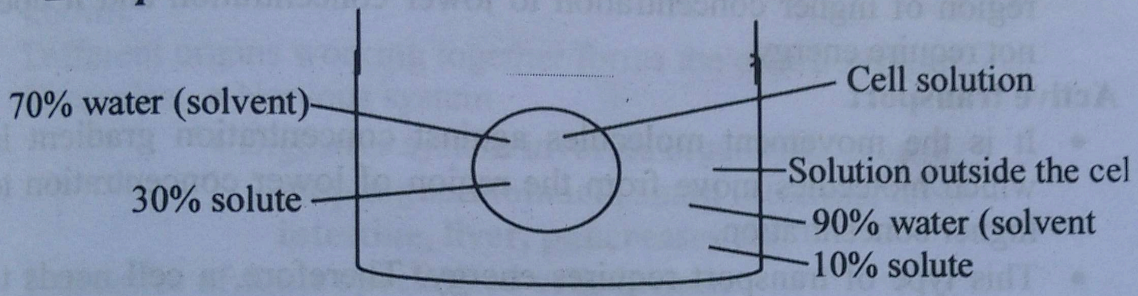


**Fig. 2.8: Hypotonic solution**

- In this case the solution outside the cell has high concentration of water than the cell contents. Therefore, because of osmosis water molecules move from the cell environment which has high water concentration in to the cell where there is low water concentration.
- In this condition the cell gains water and it swells up.

**Turgor Pressure and Plasmolysis**

If a cell is placed in hypotonic condition, the cell gains water and swell up. The swelling up of the cell creates a pressure that pushes the cell wall out wards. This pressure is called **turgor pressure**.

- To avoid bursting of the cell by turgor pressure, the cell wall in plant cells creates a **wall pressure** against turgor pressure. This is important because it keeps the stems and leaves of the plant rigid and firm.

What do you think will happen if we place an animal cell in a hypotonic solution?

If we place animal cells in hypotonic solution they will burst, since they have no cell wall and large vacuole that can hold more water.

- If a plant cell is placed in hypertonic solution it starts to lose its water contents. This results in a loss of turgor pressure and is called **plasmolysis**. This causes the vacuole to become shrink and the cell **flaccid** (becomes less rigid).
- If animal cells are placed in hypertonic solution, they will lose their water content and shrivel up. Therefore animals need homeostatic condition.

**Note:** Homeostasis means maintaining constant internal environment.

- Diffusion and osmosis are called passive transport systems. **Passive transport** requires movement of molecules from the region of higher concentration to lower concentration and it does not require energy.

**Active transport**

- It is the movement molecules against concentration gradient in which molecules move from the region of lower concentration to higher concentration.
- This type of transport requires energy. Therefore, a cell needs to expend energy to transport materials during active transport.
- Active transport is important for:
  - Absorption of mineral ions by the root hair cells from the soil. This occurs because the mineral ions are at higher concentration inside the root cells, while the mineral ions in the soil is relatively at low concentration.
  - **Translocation** – is the movement of organic materials prepared in the leaf by photosynthesis to different parts of a plant (stem root).
    - The parts like stem and root have high storage of organic materials than that of the leaf. However, organic materials move by active transport from their lower concentration leaves to higher concentrated areas like stem and root.

**Cell specialization in Humans**

- Cells are the functional units of life. This statement indicates that each cell in multicellular organisms is adapted to carry out specific functions of the body. Such cells are called **specialized cells.**
- Specialized cells are those cells which are differentiated to carry out specific activities.

- **Embryonic stem cells** are cells from an early embryo that consists of unspecialized (undifferentiated) cells that have a potential to form any type of specialized cells.

> **NB. Embryo** consists of early cells which are formed as a result of fertilization between sperm and egg cells.

- **Undifferentiated cells** are cells that are not yet specialized to perform specific functions.
- Specialized cells together form a **tissue**. Some examples of human tissues are: **Blood tissue, epithelial tissue, muscle tissue, nerve**.
- Tissues that work together form an **organ**. Among many organs found in humans' **heart, kidneys, lungs** are the most commonly known.
- Different organs working together forms the **organ system**

Examples: • Nervous system
• Digestive system involves organs like **mouth, esophagus, stomach, small intestine, large intestine, liver, pancreases** etc.

**Examples of some specialized cells include**

**a) Nerve cells (Neurons)**

- They are cells that are specialized to carry electrical nerve impulses in the body.
- The major parts of the neuron are:
  - ***Dendrite*** – to receive impulse from neighboring neurons
  - ***Cell body***–consists major structures like nucleus and mitochondria.
  - ***Axon*** – to carry nerve impulses a long distance.
  - ***Mylensheath*** – a fatty insulating layer of the axon.

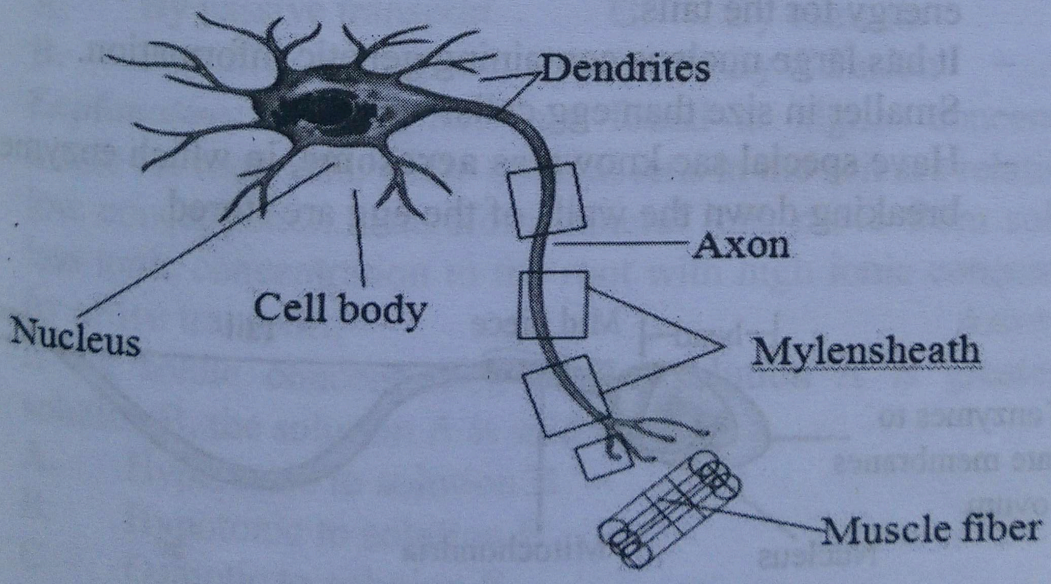


**Fig. 2.9: A neuron (nerve cell)**

### b) Reproductive cells (egg and sperm)

Reproductive cells are involved in the process of reproduction. The are:

- **Ova** or **egg** – is female reproductive cell which is found ovaries.
  - Contain half the chromosome number of the norm body cells.
  - Have large nucleus containing genetic informatio
  - Have protective coat
  - Food is stored in their cytoplasm for embryo development.

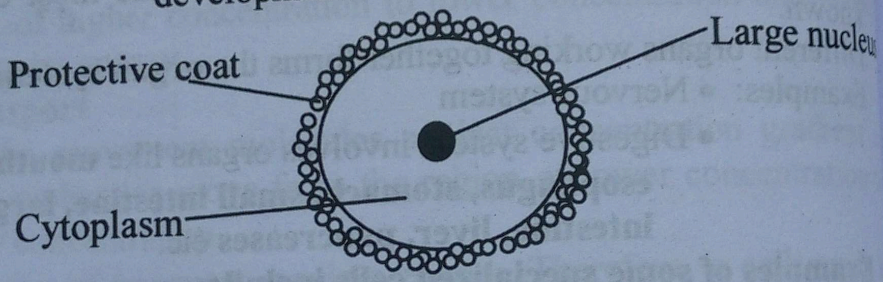


**Fig. 2.10 egg cell**

- **Sperm** – is male sex cell
  - It also contains half the chromosome number of the norm body cell.
  - Have tails to swim towards egg in female reproducti organ.
  - In middle section, they are full of mitochondria to provi energy for the tails.
  - It has large nucleus containing genetic information.
  - Smaller in size than egg cells.
  - Have special sac known as **acrosome**, in which enzymes f breaking down the walls of the egg are stored.

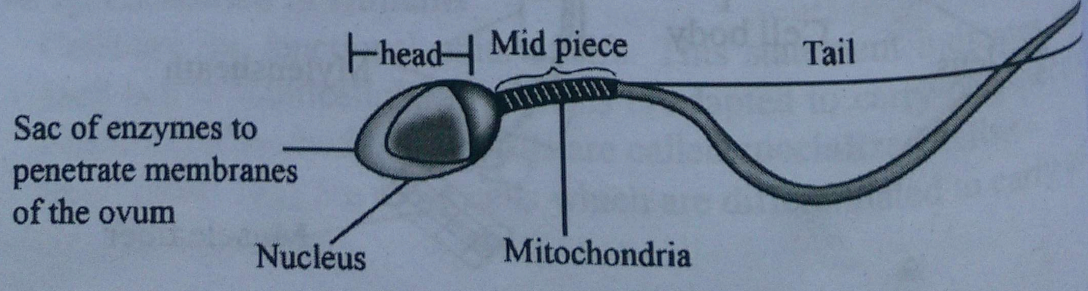


**Fig. 2.11 Sperm**



### c) Epithelial

Epithelial cells are ce internal and external are made up of epithel

- Skin, gut,

The main job of epith

- Protect tis
- Allow di another.
- Increase **microvil**

### d) Muscle

- Muscle cells body.
- Muscle cells
- They contai These protei
- Muscle cells more energy

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    D. Iso

### c) Epithelial cells

Epithelial cells are cells that form a one cell thick tissue that cover internal and external surfaces of our body. Some of the structures that are made up of epithelial cells include:

- Skin, gut, respiratory system, reproductive system etc.

The main job of epithelial cells in the body is to:

- Protect tissues underneath from damage or infection,
- Allow diffusion of materials from one area of the body to another. ***Examples***: Alveoli, gut etc.
- Increase surface area of the cell by hair like projects called **microvillus**.

### d) Muscle cells

- Muscle cells make our muscles which allow the movement of the body.
- Muscle cells have ability to **contract** and **relax**.
- They contain two types of proteins called **actin** and **myosin**. These proteins make our muscles to contract and relax.
- Muscle cells contain large number of mitochondria to provide more energy for muscular movements.

## Illustrative Example

51. Usually mineral ions in the soil are found in very dilute solutions than the solutions in plant cell. If so, how do plants absorb mineral ions needed by plant cells?

A. By passive transport C. By osmosis
B. By active transport D. By diffusion

***Explanation:*** Mineral ions are found at higher concentration inside the root cells, while mineral ions in the soil are relatively at low concentration. Therefore mineral ions move from soil with less ionic concentration to the root with high ionic concentration by active transport. **Answer: B**

52. If the solute concentration of the solution A is greater than solution B, the solution A is said to be:

A. Hypertonic to solution B
B. Hypotonic to solution B
C. Osmotic to solution B
D. Isotonic to solution B

***Explanation:*** If a cell is placed in a concentrated (strong) solution, it loses water by osmosis. If a cell is placed in hypertonic solution it loses its water content. **Answer: A**

Table below shows four fresh potato cubes (1, 2, 3 and 4) which were equal weight initially and final weight after being kept in solutions of different concentration for an hour.

| Potato cubes | Initial weight | Final weight |
|---|---|---|
| 1 | 18 grams | 18grams |
| 2 | 18 grams | 10grams |
| 3 | 18 grams | 28grams |
| 4 | 18 grams | 14grams |

53. Which of the potato cubes was kept in the most concentrated solution?

A. 3 B. 2 C. 4 D. 1

***Explanation:*** Osmosis is the movement of water molecules from dilute solution to concentrated solution. Thus the solution which is putted in concentrated solution loses more water.

**Answer: B**

54. Suppose a cell was kept in a solution of unknown concentration and found to have lost its water content after an hour. Which of the following is correct explanation about the solution and the cell? The:

A. cell is hypertonic to the solution
B. cell is isotonic with the solution
C. solution is hypotonic to cell
D. solution is hypertonic to the cell

***Explanation:***

If a cell is placed in hypertonic solution it loses, its water content and shrinks.

**Answer: D**

55. If a given cell is isotonic to 20% sucrose solution, then which one of the following solutions can be hypertonic to the cells solution?

A. 10% sucrose solution C. 20% sucrose solution
B. 15% sucrose solution D. 25% sucrose solution

***Explanation:*** If a cell is isotonic to 20% sucrose solution, since the cell and its external surrounding have the same solution. The contraction of the cell will be 20% sucrose solution, which can also means 80% water concentration.

56. If a cell is placed in 25% sucrose solution, which means 75% water concentration. The water moves from the cell to its environment by osmosis. The loss of water from a cell creates hypertonic solution.

**Answer: D**

If you open a bottled perfume and leave it for sometime in one corner of a classroom

A. You will smell the perfume better if you are away from it.
B. You will smell the perfume better if you are close to the bottle.
C. The smell is due to diffusion of the particles of the perfume
D. A and C

*Explanation*: Diffusion is the movement of molecules from the region of higher concentration to lower concentrated area, until the molecules tend to distribute uniformly.

**Answer: C**

57. If onion epidermal cells are immersed in 1M sucrose solution,
A. The solution is said to be hypertonic to the cells
B. The cells plasmolyse
C. Water molecules leave the cell
D. The cell membrane shrinks
E. All are possible answers

*Explanation*:
Onion epidermal cells are less concentrated than 1M sucrose solution. Thus, the water moves from Onion epidermal cell (high water concentration) to sucrose solution (less water concentration) by osmosis. Therefore, the sucrose solution is said to be hypertonic to the epidermal cell.

**Answer: E**

58. How do you define osmosis in a semi-permeable membrane?
A. The diffusion of water from more concentrated to less concentrated
B. The diffusion of water from less concentrated solution to more concentrated solution.
C. The diffusion of solutes from more concentrated to less concentrated solution.
D. The diffusion of solutes from less concentrated solution to more concentrated solution.

***Explanation***: Osmosis is special kind of diffusion in which the water molecules moves from dilute (less concentrated) solution to high concentrated solution, through a semi permeable membrane.

**Note:** **Dilute solution** has relatively high water and less solute concentration

**Concentrated solution** has relatively more solute concentration than the dilute (weak) solution.

59. Suppose 10% sucrose solution is isotonic to the protoplasm of a potato cylinder, the cylinder would gain water most if it is kept in a vessel containing

**Answer: B**

A. No liquid
B. Distilled water
C. 5% sucrose solution
D. 10% sucrose solution
E. 50% sucrose solution

***Explanation***: If 10% sucrose solution is isotonic to the potato cylinder. The cylinder has also 10% sucrose solution.

- 5% sucrose solution is hypotonic to cylinder which has 10% sucrose solution. Thus, since 5% sucrose solution has more water than the cylinder (10% sucrose), the water moves from the 5% sucrose solution to the cylinder (10% sucrose solution) by osmosis. As a result the cylinder gains water.
- 10% sucrose solution is isotonic to the cylinder.
- 50% sucrose solution is hypertonic to the cylinder.

60. Suppose the content of potato cells is isotonic to 20% sugar solution, in which of the following concentration of sugar solutions would a potato cube gain weight most if kept in the solution?

**Answer: C**

A. 2% B. 10% C. 20% D. 40%

***Explanation:***

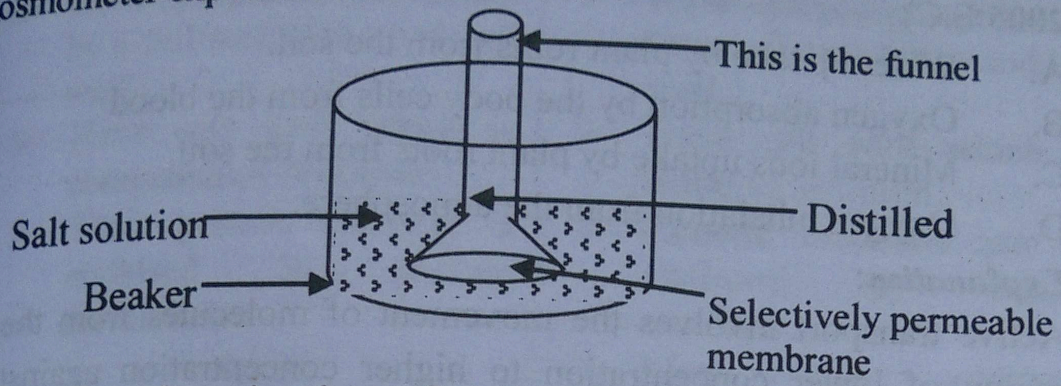
- If 20% sugar solution is isotonic to potato cells. The potato cells have also 20% sugar solution.

**Note:** Isotonic is a solution in which both solutions outside and inside the cell are the same.

- 2% sugar solution is the most dilute solution among those given in the alternation. Thus, if the potato cell which have 10% sugar solution is placed in a solution, it gains more water from the most dilute solutions.

**Answer: A**

Item number 55 is based on the diagram below which shows a simple osmometer experiment.




61. Based on the above diagram which of the following will happen after sometime? The level of:
    A. Salt solution in the beaker will decrease
    B. Water in the funnel will decrease
    C. Both the water and the salt solution remain the same
    D. Water in the funnel will increase

***Explanation:***

The water moves from dilute solution (distilled water) to the concentrated solution (salt solution) by osmosis. Therefore, funnel which consists of distilled water loses water.

**Answer: B**

62. Which of the following organelles is correctly matched to its function? (EGSECE, 2005 E.C)
    A. Cytoplasm – barrier around outside of the cell
    B. Cell membrane – controls all activities of the cell
    C. Nucleus – site of protein synthesis
    D. Mitochondrion – power house of the cell

***Explanation:***

- **Cytoplasm** – is the part which is filled with jelly like substance outside the nucleus. It also consists of organelles of a cell.
- **Cell membrane** – It a lipo protein layer which controls the movement of substances in and out of a cell.
- **Nucleus** – Is part which controls and directs the activities of a cell.
- **Mitochondria** – is the site of cellular respiration. It is also called 'a power house of a cell.

**Answer: D**

63. Which of the following is an active transport process? (EGSECE, 2005 E.C)

A. Water up take by plant roots from the soil.
B. Oxygen absorption by the body cells from the blood
C. Mineral ions uptake by plant roots from the soil.
D. Oxygen inhalation from the atmosphere.

***Explanation:***

Active transport involves the movement of molecules from the region of lower concentration to higher concentration against concentration gradient.

- Some examples of active transport are:
  - Uptake of mineral ions by plant roots from the soil.
  - Translocation – movement of organic substances from the leaf to different parts of a plant.

**Answer: C**

64. Which of the following organelles is found in large number in cells that carryout active transport process frequently? (EGSECE, 2005 E.C)

A. Ribosome
B. Nucleus
C. Mitochondria
D. cytoplasm

***Explanation:***

- *Active transport system* requires expenditure of energy. Thus, the energy synthesizing organelle of a cell is called **mitochondria.**

**Answer: C**

65. Which of the following is a similarity between a bacterial and plant cells?

A. The presence of cytoplasm surrounded by a membrane and cell wall.
B. The enclosed genetic material in nuclear membrane.
C. The material out of which the cell wall is made.
D. Their possession of flagella and capsule.

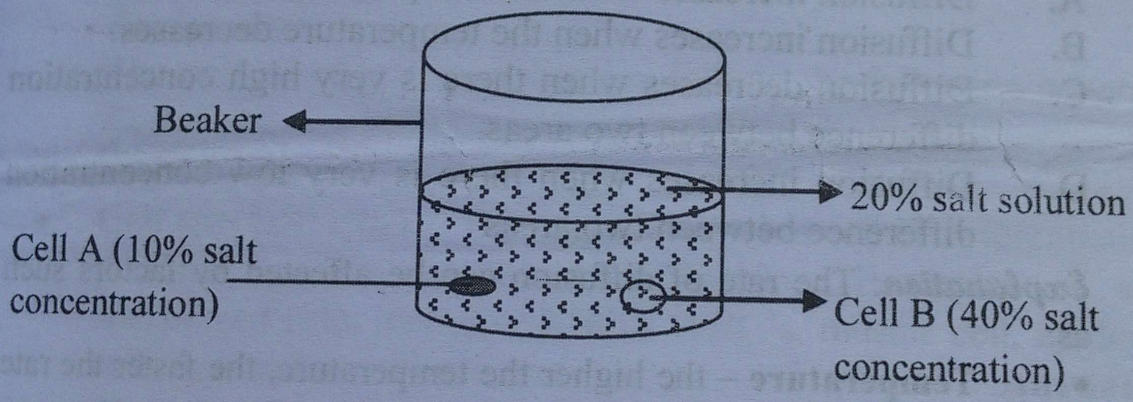
***Explanation:***

- Comparison between plant cell and bacterial cell

| Plant cell | Bacterial cell |
|---|---|
| ➢ Have cytoplasm surrounded by a cell membrane and cell wall<br>➢ Have cell wall which is composed of cellulose<br>➢ Their genetic material is enclosed in nuclear membrane | ➢ Have cytoplasm surrounded by a cell membrane and cell wall.<br>➢ Have cell wall which us made up of chitin.<br>➢ Genetic materials are not enclosed by nuclear membrane<br>➢ May have additional Features like flagella and capsule. |

**Answer: A**

66. This item is based on the following diagram which shows two cells of different concentration immersed in a beaker containing 20% salt solution.




What will be happen in the cells after two hours?

A. Cell A increases in mass.
B. Net water moves in to cell B.
C. Net water moves in to cell A.
D. Cell B decreases in mass.

*Explanation*:

- Cell 'B' contains less water than its environment. Therefore, it gains water from its environment by osmosis. Thus cell 'B' increases in mass.
- Cell 'A' contains more water than its environment. Therefore, it loses water to its environment by osmosis. Thus, cell 'A' decreases in mass.

- Osmosis is the movement of water molecule from its higher concentration to laver concentration through semi permeable membranes.

**Answer: B**

67. If an animal cell is put in concentrated salt solution, which of the following event will happen? (EGSECE, 2006 E.C)
   A. The cell becomes plasmolyzed
   B. Water moves in to the cell.
   C. The cell becomes shrunk
   D. Salt moves in to the cell,

   ***Explanation***: If an animal cell is put in a concentrated (strong) solution, it loses its water content by osmosis. Thus, it become shrunk.

**Answer: C**

68. Which of the following is true about the rate of diffusion? (EGSECE, 2006 E.C)
   A. Diffusion increases when the temperature increases.
   B. Diffusion increases when the temperature decreases.
   C. Diffusion decreases when there is very high concentration difference between two areas.
   D. Diffusion increases when there is very low concentration difference between two areas

   ***Explanation***: The rate of diffusion can be affected by factors such as:

   - **Temperature** – the higher the temperature, the faster the rate of diffusion.
   - **Concentration gradient** – the higher the concentration difference between two areas, the faster the rate of diffusion.

**Answer: A**

69. A special type of movement of water through partially permeable membrane across concentration gradient is called;
   A. Diffusion
   B. Passive transport
   C. active transport
   D. osmosis

   ***Explanation***:

   - **Diffusion** – is the movement of molecules from the region of higher concentration to less concentration until it tends to distribute uniformly
   - **Osmosis** – is the movement of water molecule from its regio of higher concentration to lower concentration throug semi-permeable membrane.

- **Passive transport** – Is a movement of substances from the region of higher concentration to lower concentration along a concentration gradient.
  - It does not expend energy
  - Examples: Diffusion, osmosis
- **Active transport** – Is the movement of molecules from a region of lower concentration to higher concentration (against concentration gradient).
  - It expends energy

**Answer: D**

70. The main purpose of cell specialization in multicellular organisms is:
A. to increase in size
B. to pass genetic information
C. to facilitate reproduction
D. to perform a particular function

***Explanation***:
The process of cell specialization follows the following pattern
*Cell division → cell elongation → cell maturation → cell specialization*

- **Cell specialization** – The stage in which a cell is differentiated to perform specific or particular function. Examples of some specialized cells includes; RBC's WBC's, muscle cell, epidermal cell, liver cells, neurons, etc. **Answer: D**

71. This item is based on the table below which shows the comparison between plant and animal cells. (EGSECE, 2006 E.C)

| No. | Animal cell | Plant cell |
|---|---|---|
| 1 | Large vacuole | Small vacuole |
| 2 | Cell membrane present | Cell membrane absent |
| 3 | Cell wall present | Cell wall absent |
| 4 | Chloroplast absent | Chloroplast present |

Which of the following comparison in the above table is correctly matched?
A. 2 B. 3 C. 1 D. 4

***Explanation***:
The correct match about comparison between plant animal cells is:

| No. | Animal cell | Plant cell |
|---|---|---|
| 1 | Small and temporary vacuole | Large, central and permanent vacuole |
| 2 | Cell membrane present | Cell membrane present |
| 3 | Cell wall absent | Cell wall present |
| 4 | Chloroplast absent | Chloroplast present |

**Answer: D**

72. Cell membrane allows some substances to pass in to the cell. It is because the cell membrane is: (EGSECE, 2006 E.C)
A. Flaccid
B. Strong
C. selectively permeable
D. permeable

*Explanation*: Cell membrane is a selectively/partially or semi-permeable membrane that controls the movement of substances in and out of the cell

**Answer: C**

## Post Test 5

***Part I: Choose the best option from the given alternatives.***

1. If a plant cell is placed in a hypotonic solution it will:
A. Burst
B. Flaccid
C. Shrink
D. Plasmolyse
2. Which one of the transportation mechanism in a cell is against concentration gradient?
A. Osmosis
B. Active transport
C. Diffusion
D. Passive transport
3. Which of the following is not specialized cell?
A. Sperm cell
B. Ovum
C. Stem cell
D. Neuron
4. Which one of the following is the correct sequence of chain of cell specialization?
A. Cell division → cell elongation→ cell maturation→ cell specialization
B. Cell elongation → cell division→ cell specialization → cell maturation
C. Cell maturation→ cell division → cell specialization → cell elongation
D. Cell specialization → cell maturation → cell elongation → cell division
5. During cell specialization which one comes first?
A. Cell maturation
B. Cell specialization
C. Cell elongation
D. Cell division



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C. 20%
D. 25%

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6. The movement of water molecules along concentration gradient across a cell membrane is:
   A. Diffusion C. Osmosis
   B. Active transport D. Facilitated diffusion
7. The bursting of a plant cell during turgor pressure in plant cell is prevented by:
   A. Cell membrane C. Nucleus
   B. Cell wall D. Chloroplast
8. The solution in and outside a cell is uniform in which of the following solution?
   A. Hypertonic solution C. Isotonic solution
   B. Hypotonic solution D. None of the above
9. Suppose a cell was kept in a solution of unknown concentration and found to have stayed the same after an hour. Which of the following is correct explanation about the solution and the cell? The:
   A. cell is hypertonic to the solution
   B. cell is isotonic with the solution
   C. solution is hypotonic to cell
   D. solution is hypertonic to the cell
10. If a given cell is placed in 20% sucrose solution, then which one of the following solutions can be hypertonic to the cells solution?
   A. 10% sucrose solution
   B. 15% sucrose solution
   C. 20% sucrose solution
   D. 25% sucrose solution

## Supplementary Exercise on Unit Two

1. Suppose an object is magnified by using ocular of 10x and objective lens of 10x. If the magnified size of an object is 2mm, what was the original size of an object in mm?
   A. 2mm B. 0.2mm C. 0.02mm D. 20mm
2. Some object is seen under the microscope and its size is increased by 100x. This statement explains which ability of the microscope?
   A. Mounting C. staining
   B. Resolution D. Magnification

20. The part of a plant cell where starch is stored is
    A. Chloroplast
    B. Chromoplast
    C. leucoplast
    D. Centriole
21. Which of the following organelles are common to both plant and animal cell?
    A. Mitochondria and chloroplast
    B. Golgi body and centriole
    C. Lysosome and ribosome
    D. Mitochondria and Ribosome
22. An organelle of a cell that is involved in protein synthesis is
    A. Mitochondria
    B. Golgi body
    C. Ribosome
    D. lysosome
23. Plasmolysis takes place when a cell is kept in
    A. Isotonic environment
    B. Hypertonic environment
    C. Hypotonic solution
    D. None
24. Which of the following solutions of sucrose is hypotonic to all the rest?
    A. 15% B. 20% C. 50% D. 80%
25. Which of the following is due to the loss of water from cells?
    A. Brusting of cells
    B. Osmosis
    C. Turgor pressure
    D. wilting
26. If some cell is placed in hypotonic solution, it will
    A. Shrink
    B. Swell up
    C. plasmolysed
    D. remain unchanged
27. Specialized cells are best described as
    A. Cells that have a potential to differentiate in to different cells.
    B. Cells that are found at the embryo.
    C. Cells which are adapted to carry out specific functions
    D. Cells which divide mitotically
28. Which of the following involves passive transport system?
    A. Absorption water by the root hair cells.
    B. Movement of organic materials from the leaf to different parts of the plant
    C. Absorption of mineral ions by the root hair cells from the soil
    D. None of the above

29. Which of the following solutions of sucrose is hypertonic to all the rest?
A. 15% B. 20% C. 50% D. 2%

30. Which of the following is found in animal cells but not in plant cells?
A. Mitochondria C. lysosome
B. Chloroplast D. cytoplasm

31. Enzymes concerned with the release of energy from food are mostly found in the cellular structures named:
A. Nuclei C. cell membrane
B. Mitochondria D. Ribosome

Items numbers 32 to 34 are based on the following experiment. Pieces of potato cylinders measuring 20grams in weight were immersed in different solutions and after 1 hour the following result were obtained as given in table below.

| Potato | Initial weight (g) | Final weight (g) |
|---|---|---|
| Water | 20 | 30 |
| 20% sucrose | 20 | 15 |
| 50% sucrose | 20 | 10 |

32. The potato cylinders gained weight when immersed in water because:
A. Water entered the potato cells by osmosis.
B. Water entered the cells along concentration gradient of water
C. Water moved against its gradient
D. A and B

33. The potato cylinders lost weight when immersed in 50% sucrose solution because:
A. Water left the potato cells following its osmotic gradient
B. Sucrose molecules entered the potato cells.
C. The potato cells shrinked by loosing water
D. Only A and C

34. What do you learn from the above experiment?
A. The principle of osmosis
B. Cell membrane is selectively permeable
C. Permeability depends on size of molecules
D. All are possible answers.

35. Given
1. Nucleus 5. Cell membrane
2. Nucleolus 6. Central vacuole

3. Mitochondria
4. Chloroplast
7. Chromosome

Structure found in animal cells but not in plant cells are:

A. 2,3,5,7 C. 4,6 only
B. 1,2,3,7 D. 3 only E. None of the above

36. The uptake of mineral ions from the soil in to the plant is by:

A. Active transport C. Diffusion
B. Osmosis D. Translocation

**III. Blank space items**

37. The ability of a microscope to reveal the fine detail of an objects call _____.
38. Proteins that are to be secreted from the cell are synthesized on _____ which are bound to the _____.
39. The folds in the inner membrane of the mitochondria are called _____.
40. In addition to have cell membrane, plant cells are surrounded by _____.

**Answers to Post Test 1**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | B | 2. | A | 3. | D | 4. | B | 5. | A |
| 6. | B | 7. | C | 8. | B | 9. | B | 10. | D |

**Answers to Post Test 2**

***Part I***: 1. C 2. A 3. B 4. C 5. B
***Part II:*** 1. E 2. B 3. G 4. F 5. C
6. D 7. A

**Answer to Post Test 3**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | B | 2. | C | 3. | A | 4. | D | 5. | C |
| 6. | A | 7. | B | 8. | B | 9. | D | 10. | D |

**Answers to Post Test 4**

***Part I***: 1. A 2.B 3. C 4. D 5. D 6. C



***Part II:***

Plan
- Cel
- Cer
- Chl

1. B
_____ A

1. C
2. D
3. C
4. B
5. C
6. B
7. B
8. B

***Part II:***

| Plant cells only | Animal cells only | Found in both |
|---|---|---|
| - Cell wall<br>- Central vacuole<br>- Chloroplast | - Centriole<br>- Lysosome | - Cell membrane<br>- Nucleus<br>- Golgi body<br>- Endoplasmic reticulum<br>- Ribosome |

**Answers to Post Test 5**

1. B 2. B 3. C 4. A 5. D 6. C 7. B 8. C 9. B 10. D

**Answers to Supplementary Exercise on Unit Two**

1. C
2. D
3. C
4. B
5. C
6. B
7. B
8. B
9. C
10. D
11. B
12. B
13. D
14. B
15. D
16. A
17. C
18. B
19. C
20. C
21. D
22. C
23. B
24. A
25. D
26. B
27. C
28. A
29. C
30. C
31. B
32. A
33. C 34. A 35.E 36.A
37. Resolution
38. Ribosome, Rough Endoplasmic reticulum.
39. Cristae
40. Cell wall

Unit – 3

# 3. HUMAN BIOLOGY AND HEALTH

## 3.1 Food and Nutrition

**Food** is anything to be eaten to provide energy for the body, provide raw materials for growth, to fight disease and maintain a healthy body.

Food contains **six major groups** of nutrients. These are

- Carbohydrate
- Protein
- Fat (lipid)
- Vitamins
- Minerals
- water

Nutrients found in a food can be grouped as:

- ***Micronutrients:*** - are groups of nutrients that are needed by the body in trace or very small amount. These are minerals and Vitamins.
- ***Macronutrients***: - are nutrients that are required by the body in large amount.

These include carbohydrate, protein, lipids.

**Nutrients** are useful substances that are found in a food.

### 1) Carbohydrates

Carbohydrates are the quickest, most readily available source of energy for the body.

- They are composed of three elements; **carbon, hydrogen** and **oxygen.**

The general formula of carbohydrate is:

$(CH_2O)n$ . . . . for $n \geq 3$

In carbohydrates the ratio of **hydrogen** and **oxygen** is 2:1

- Carbohydrates are also the storage forms of food in plants and animals.
  - **Starch** – is the storage form of carbohydrates in plants.
  - **Glycogen** – is the storage form of carbohydrate in animals.
    - It's also called ***'animal starch'.***
- Some of the food stuffs that are rich in their carbohydrate content are cereals, honey, candy, potato, spaghetti, etc.
- Carbohydrates are classified in to three major classes based on the number of sugar units which they contain. These are:
  - Monosaccharide
  - Disaccharides
  - Polysaccharides

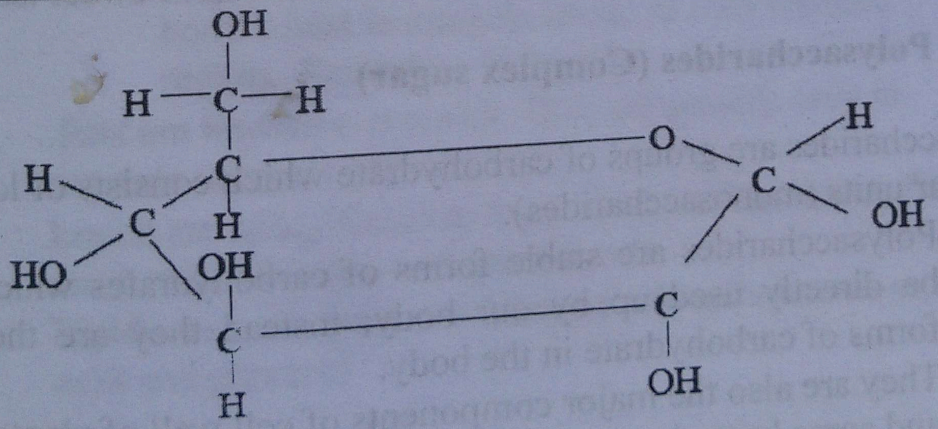
**Note:** One sugar unit contains 3 – 8 carbon atoms.

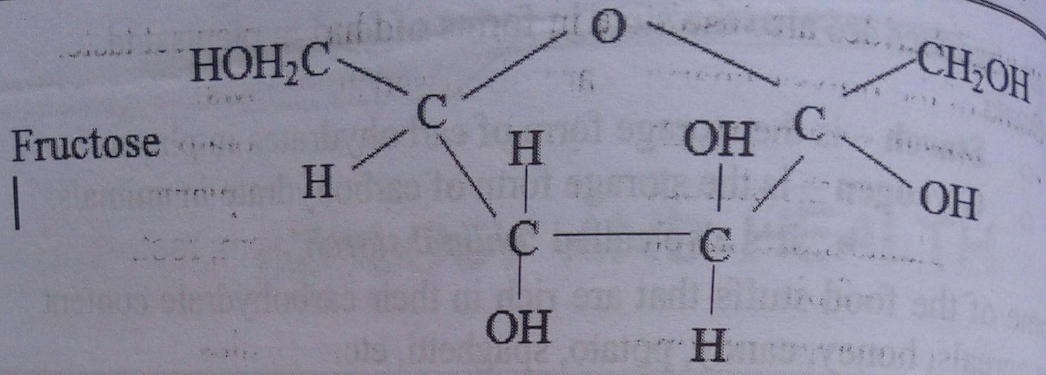
## Monosaccharide (Simple sugars)

- Monosaccharide is the simplest and soluble forms of carbohydrates.
- They are composed of one sugar unit. Sugar units are also called **saccharides.**
- The most commonly known examples of monosaccharide are, **galactose, glucose,** and **fructose**.
- Glucose, galactose and fructose have the same molecular formula which is $C_6H_{12}O_6$. However, they are structurally different. Such kinds of molecules are called **isomers**.

**Isomers** are substances that have the same molecular formula but different structural arrangement.

Glucose






*All monosaccharides are soluble in water and have sweet taste.*

**ii) Disaccharides (double sugars)**

Disaccharides are groups of carbohydrates which are composed of two sugar units. They are formed by **condensation reaction** between two monosaccharides.

***Condensation reaction** is a reaction that produces water as a biproduct.*

$$C_6H_{12}O_6 + C_6H_{12}O_6 \longrightarrow C_{12}H_{22}O_{11} + H_2O$$

Monosaccharide Monosaccaride disaccharide water

- The most commonly known examples of disaccharides are:
  - ***Sucrose*** – found in table sugar which we normally use.
  - ***Lactose*** – sugar found in milk
  - ***Maltose*** – sugars found in germinating seeds of cereals like barley.

glucose + glucose ⟶ Maltose + water
glucose + fructose ⟶ sucrose + water
glucose + galactose ⟶ lactose + water

- All disaccharides are soluble in water and have sweet taste.

**ii) Polysaccharides (Complex sugar)**

Polysaccharides are groups of carbohydrate which consists of long chain of sugar units (monosaccharides).

- Polysaccharides are stable forms of carbohydrates which cannot be directly used up by our body; instead they are the storage forms of carbohydrate in the body.
- They are also the major components of cell wall of plants, fungi and some bacteria.

Polysaccharides are insoluble in water and have no sweet taste.
Some of the polysaccharides are discussed as follows:

- ✓ ***Starch*** – found richly in food items such as potato.
  – are the storage form of carbohydrate in plants.
  - plants store starch in their parts such as stem, root, fruit.
  – composed of long chain of glucose molecules.
- ✓ ***Glycogen*** – is the storage form of carbohydrate in animals.
  – They are stored mainly in the liver and muscles.
- ✓ ***Cellulose*** – The major component of cell wall in plant cell.
- ✓ **Murein** –are components of bacterial cell wall.
- ✓ Polysaccharides are **polymers** which are composed of **monomer units** of monosaccharide by condensation reaction.

**2. Fats and Oils**

- These are groups of nutrients which are composed of elements **carbon, hydrogen** and **oxygen**. However, unlike carbohydrates in which the ratio of hydrogen and oxygen is 2:1, in Fats (lipids) there is little oxygen when compared to that of carbohydrate.

*Fats contain twice as much energy than that of carbohydrates.*

- They are important for:
  - Storage of energy in the body.
  - Structural components of cell membrane.
  - Insulation of layers under the skin.
  - Regulation of body temperature by maintaining body's heat in animals living in cold climatic region. **Example:** Polar beer
- Fats are insoluble in water. They are soluble only in **organic solvents** like alcohol.
- Lipids are richly found in food items such as butter, fatty meat, vegetable oils and ground nuts.
- Lipids are built from two kinds of sub units known as **fatty acid** and **glycerol**

### i. Glycerol

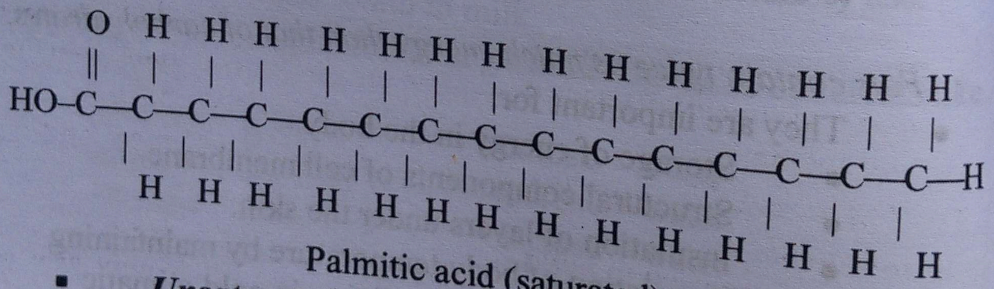
Glycerol is a three – carbon alcohol with each carbon bearing a hydroxyl (– HO) group. The three carbons form the back bone of fat molecule to which three fatty acids are attached.

- Their molecular formula is $C_3H_8O_3$.

### ii. Fatty acids

Fatty acids are composed of long hydrocarbon chains (consisting of or carbon and hydrogen atoms) ending in carboxyl (- COOH) group. Fatty acids vary in length.

- The most common groups of fatty acids are even numbered chains of 14 to 20 carbons. Accordingly, fatty acids are classified as **saturated** and **unsaturated** fatty acids.
- ***Saturated Fatty acids:*** contain single bonds between carbon atoms and contain internal carbon atoms having two hydrogen side groups.
  - Are mostly solid at room temperature
  - The more highly saturated fats are richer in energy, but they may up set the normal balance of fatty acids in the body, a situation that can lead to heart disease.
  -

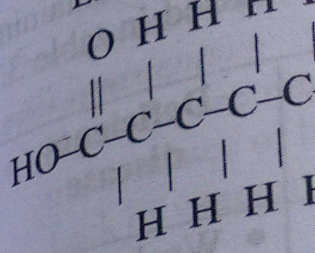


Palmitic acid (saturated)

- ***Unsaturated Fatty acids:*** – contain double bonds between carbons that replace some hydrogen ions. They contain fewer number of hydrogen atoms than saturated fatty acids.
  - Many of them are obtained from plants
  - are relatively healthier and contain less number of cholesterol.
  - Examples are, oleic acid (vegetable oil)



Linolenic acid



### 3. Protein

Proteins are organic **hydrogen, oxygen** and **sulfur**.

- Amino acids 20 different and form pro
- The bond be **link**. The **condensatio**
- There are va in **number**
- Some protei water.
- ***Water inso*** make conne and feather
- ***Water solu*** hormones.
- Proteins are
- Some food egg, fish, c

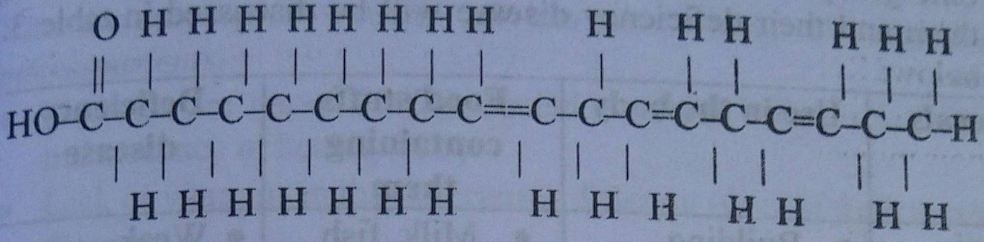
### 4. Minerals

Minerals are gro amount, and class

- Shortage of **deficie**

**Deficie**

Linolenic acid (linseed oil)



### 3. Protein

Proteins are organic substances that are composed of elements **carbon, hydrogen, oxygen** and **nitrogen**, sometimes it includes **phosphorous** and **sulfur**.

- Amino acids are the building blocks of proteins. There are about 20 different types of amino acids that are joined to one another and form protein.
- The bond between amino acids to form protein is called **peptide link**. The link between amino acids is an example of **condensation reaction**.
- There are various types of proteins which differ from one another in **number** and **sequence** of amino acids which makes them.
- Some proteins are insoluble in water, while others are soluble in water.
- ***Water insoluble proteins*** are structural proteins that are used to make connective tissues, tendons, bone matrix, hair, nail, horns and feathers.
- ***Water soluble proteins*** form antibodies, enzymes and some hormones.
- Proteins are used to build and repair the body tissues.
- Some food stuffs that are rich in their protein content are; milk, egg, fish, chicken meat.

### 4. Minerals

Minerals are group of nutrients are need by the body in a very small amount, and classified as **micronutrients**.

- Shortage or lack of minerals in the diet may result in some kind of **deficiency disease.**

> ***Deficiency disease*** is a disease that is caused by a shortage of nutrients in the food or diet.